From Reproduction to Generalisation: Pseudo-Processive Kinetics with a Fingerprint-Informed Prior (PPK-FIP) for the Lipase-Catalysed Esterification of Trimethylolpropane

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Abstract

Kinetic models of the lipase-catalysed esterification of trimethylolpropane (TMP) with fatty acids, a route to fully bio-based biolubricants, must describe three competing Ping Pong bi bi reactions with few, correlated parameters. Nishimura et al. (*Biomass* 2026, 6, 12) proposed a two-step semi-mechanistic method that first estimates the enzyme selectivities against the degree of reaction ("fingerprinting") and then fixes them while fitting lumped constants against time. Because every data point of that study is public, we could put one chain of questions to identical data. (1) Is the study reproducible? All 36 data checks and all fingerprint fits were reproduced exactly; several printed dynamic fits, however, were not at their optimum, the full model is structurally non-identifiable, and one case study could not be reproduced, so in-sample fit statistics alone are an unreliable guide to the model. (2) What does the remaining misfit point to? Structured residuals indicate pseudo-processivity (re-acylation of an ester before it leaves the active site) in the mass-transfer-limited data and, in one profile, first-order enzyme deactivation. (3) Can these mechanisms be added without losing the stabilising fingerprint information? PPK-FIP (pseudo-processive kinetics with a fingerprint-informed prior) treats the fingerprint estimate as a calibrated Bayesian prior rather than a fixed value, averages model forms by BIC, selects hyper-parameters by leave-one-case-out meta-validation, and uses an exact reparameterisation of the shared Ping Pong denominator that makes exhaustive validation cheap. (4) Does it describe and predict better? PPK-FIP lowered the minimised sum of squares in all six fits (by 3.7–73.0%) and, in leave-one-time-point-out cross-validation with both steps of the original method re-estimated in every fold, reduced the prediction error by 16.2% relative to the original method and by 8.2% relative to its full variant (both $p < 0.001$). Ablation traced the gain to pseudo-processivity and to the prior, with within-batch deactivation acting specifically in the 100 °C profile. (5) Are the mechanisms genuine? The processive composition path transferred across water activities, and Monte-Carlo experiments showed that the two-step method overestimates the selectivity S_M by 76% when pseudo-processivity is present, whereas PPK-FIP is nearly unbiased and does not invent processivity when it is absent. (6) Is mechanism still needed? Gaussian processes, MLP ensembles, the pretrained TabICL foundation model and neural ODEs interpolated densely sampled profiles but failed in extrapolation; a Gaussian-process discrepancy added to PPK-FIP (PPK-FIP-GP) gave the lowest interpolation error of all methods (49.1% below the original method) while extrapolating like the mechanistic model. The fingerprint idea is therefore most valuable as prior information, pseudo-processivity should be modelled rather than absorbed into apparent selectivities, and kinetic parameters should be reported with out-of-sample validation.

Keywords: Ping Pong bi bi; lipase; trimethylolpropane; biolubricant; kinetic model; selectivity; pseudo-processivity; enzyme deactivation; Bayesian prior; cross-validation; reproducibility; hybrid modelling

1 Introduction

Lipases catalyse esterifications, transesterifications and hydrolyses that are central to the valorisation of biomass-derived lipids [1–3]. Many of these processes are complex in the kinetic sense: several substrates compete for the enzyme, or the product of one reaction is the substrate of the next. The esterification of trimethylolpropane (TMP) with fatty acids, a key step towards fully bio-based biolubricants because both TMP and the fatty acids can be produced from biomass [4, 5], is a prototypical case. TMP (Z) is converted to the monoester (M), the diester (D) and the triester (T) in three consecutive reactions, each of which follows the Ping Pong bi bi mechanism [6, 7].

Reactor design and optimisation require kinetic models of such processes, but fully mechanistic Ping Pong rate equations contain many correlated constants that cannot be estimated reliably from concentration–time profiles [8, 9]. Nishimura et al. [10] recently proposed in *Biomass* a two-step semi-mechanistic approach that

circumvents this problem. In the first step the selectivities of the enzyme for the second and third esterification (S_M, S_D) are estimated with an "irreversible fingerprinting model" in which the degree of reaction c replaces time as the independent variable, so that enzyme saturation, inhibition and denaturation cancel out of the equations [11–13]. In the second step S_M and S_D are fixed and a dynamic model with lumped empirical constants (k_A, F_A, F_{AZ}, F_{AM}, F_{AD}) is fitted against time. Four literature case studies [9, 14–16] were described "well", judged by the minimised sum of squares F_{obj} and the Bayesian information criterion (BIC).

The approach is attractive because it is transparent and because every data point is published in the Supplementary Material. Complete public data are rare in kinetic modelling and offer an opportunity that we take up here: the method can be audited, reproduced to the last printed digit and improved on exactly the same data. Three features of the original analysis motivate this study, and they are connected. (i) Pseudo-processivity, the re-acylation of a freshly formed ester before it leaves the active site, which mass-transfer limitation produces in the data of Bornadel et al. [9, 13], was deliberately excluded, so the selectivities of that case are, in the authors' own words, only "apparent". (ii) Enzyme deactivation was handled by a separate rate constant for each reuse batch, never within a batch. (iii) The models were judged only in-sample, on the data used for fitting, and because S_M and S_D are treated as known in step 2, the two parameters estimated from the same data in step 1 are absent from the reported BIC. The connection is this: the two-step method buys its stability by fixing the selectivities from the c-domain; if the c-domain model omits a mechanism, the resulting error is locked into step 2, and an in-sample criterion applied to step 2 alone cannot reveal it.

This observation defines the main line of the paper, which we state as a single argument and then test step by step (Table 1). First, the published analysis is reproduced from the public data and audited, which shows what in-sample statistics can and cannot establish. Second, the misfit left by the two-step model is examined for its mechanistic signature. Third, we introduce **PPK-FIP** (*pseudo-processive kinetics with a fingerprint-informed prior*), which keeps the Ping Pong core and the central idea of Nishimura et al., that selectivities are best learned against the degree of reaction, but (a) adds pseudo-processivity and first-order deactivation as explicit, testable mechanisms, (b) replaces the hard "fix-then-fit" constraint by a calibrated Bayesian prior centred on the fingerprint estimate, (c) averages the simplified and full lumped-constant forms with BIC weights, choosing the prior hyper-parameters by leave-one-case-out meta-validation, never with the target data, and (d) exploits an exact reparameterisation of the shared Ping Pong denominator that makes global optimisation and exhaustive cross-validation affordable. Fourth, PPK-FIP is compared with the original method on the paper's own criteria and, decisively, out-of-sample, and an ablation attributes the gain to its components. Fifth, cross-condition transfer and Monte-Carlo experiments with known truth test whether the added mechanisms are genuine properties of the reaction systems rather than extra flexibility. Sixth, the mechanistic estimators are set against modern black-box learners (Gaussian processes, MLP ensembles, the pretrained tabular foundation model TabICL [17] and neural ordinary differential equations [18]) and then combined with them in a hybrid, PPK-FIP-GP. Every figure and table of the paper belongs to one of these six steps, and after each result we state what it contributes to the argument.

Table 1: Main line of the study: six questions, the experiments that answer them, and where the evidence is shown. Figures 1, 2 and 3 (Section 2) describe the method and the data that all six steps share.

Step	Question	Experiment	Evidence
1	Is the published analysis reproducible from its public data, and is in-sample fit a reliable guide to the model?	Data audit; re-fit of both steps; multistart and profile-likelihood analysis	Tables 2–5; Figures 4, 5
2	Which mechanisms does the remaining misfit point to?	Generalised model fitted to the full data; residual and composition-path diagnostics	Tables 6, 7; Figures 7–9
3	Does the generalised model predict unseen data, not merely describe seen data?	Leave-one-time-point-out cross-validation; forward extrapolation	Tables 8, 9; Figures 10–12
4	Which component carries the gain, and must the fingerprint be used as a prior?	Component ablation; prior width; multi-task pooling	Tables 10, 11; Figures 13, 14
5	Are the added mechanisms genuine properties of the reaction systems?	Cross-condition transfer; Monte-Carlo truth recovery	Tables 12, 13; Figures 15, 16
6	Is mechanistic structure still needed, and how should machine learning be used?	Black-box baselines; Gaussian-process discrepancy hybrid	Tables 14–16; Figures 17, 18

2 Materials and Methods

2.1 Overview of the Two Estimators

Figure 1 places the original method and PPK-FIP side by side. In the original two-step method (panel a) the profiles are first plotted against the degree of reaction c and fitted with the irreversible fingerprint model, which yields S_M and S_D; these are then held fixed while the dynamic model is fitted against time in one local search, and the "simplified" model is chosen by comparing in-sample F_{obj} and BIC. In PPK-FIP (panel b) the composition path in c is generalised to include pseudo-processivity, an exact time map adds the lumped constants and within-batch deactivation, and a single maximum-a-posteriori (MAP) fit estimates all parameters at once from global multistarts; the fingerprint estimate is not discarded but enters as the centre of a calibrated prior (dashed arrow), while shrinkage priors keep the new mechanisms at zero unless the data demand them. Two model forms are then averaged with BIC weights, and the prior hyper-parameters are chosen by meta-validation across the other case studies. The three blocks of panel (b) answer, in turn, the three open points of the Introduction: the generalised path addresses pseudo-processivity (i), the time map addresses within-batch deactivation (ii), and the prior and the out-of-sample protocols in the bottom box address the fix-then-fit constraint and the in-sample-only assessment (iii). Both estimators feed the same evaluation battery, which implements Steps 1–6 of Table 1.

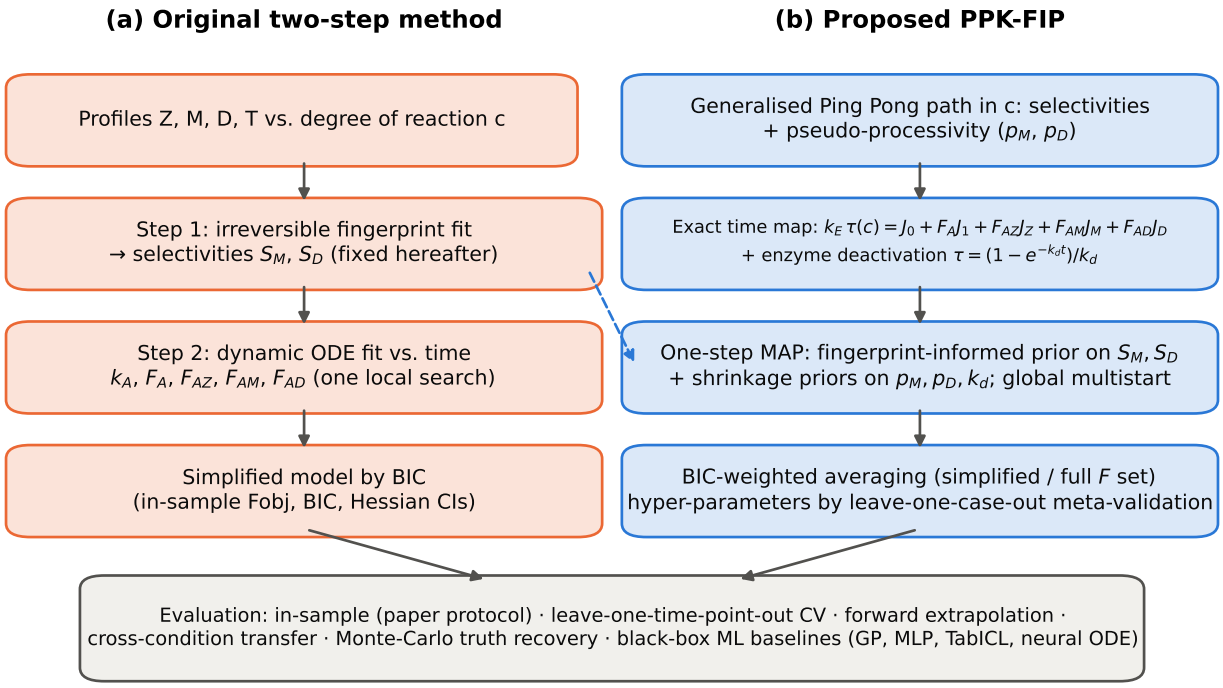


Figure 1: Original two-step method (a) and the proposed PPK-FIP (b). The dashed arrow indicates that the fingerprint estimate, a fixed value in (a), becomes the centre of a calibrated prior in (b). Both estimators are evaluated with the same protocols (bottom).

2.2 Data and Audit Protocol

All reaction profiles were taken programmatically (no manual typing) from Tables S1–S6 and S17–S21 of the Supplementary Material of [10]: Åkerman et al. at 70 and 100 °C (oleic acid, Novozym 435, 3:1) [14]; the first experiment of Bornadel et al. (same system) [9]; Tao et al. at water activities 0.25, 0.35 and 0.45 (caprylic acid, lipase of *Candida* sp. 99–125, 3.2:1) [15]; and the first three reuse batches of Mao et al. (pelargonic acid, same lipase) [16]. Species are molar percentages relative to the initial TMP, and $c = M + 2D + 3T$ is the percentage degree of reaction. In total there are 198 rows in 11 profiles, of which the 9 profiles used in the paper contain 533 fitted observations (Table 2).

Figure 2 shows the nine fitted profiles against time (left panel of each pair) and against c (right panel). Three features of the data recur throughout the paper. First, the sampling density differs by a factor of four between studies, from 7–9 sampling times in the Åkerman profiles to 30 in Bornadel; the sparse Åkerman profiles are the ones in which prior information matters most (Step 4). Second, against c the three Tao water activities nearly coincide, and so do the three Mao batches, although their time-scales differ: this is the

Table 2: Reaction profiles used (Supplementary Material of [10]); n_t = sampling times after $t = 0$.

Profile	Table	System	n_t	$t_{\max}$ (h)	Species
Åkerman 70	S1	oleic, CALB, 70 °C, 3:1	9	64	Z,M,D,T
Åkerman 100	S2	oleic, CALB, 100 °C, 3:1	7	24	Z,M,D,T
Bornadel	S3	oleic, CALB, 3:1	30	24	Z,M,D,T
Tao025	S4	caprylic, *Candida* sp., a_w 0.25, 3.2:1	16	75	Z,M,D,T
Tao035	S5	caprylic, a_w 0.35	18	79	Z,M,D,T
Tao045	S6	caprylic, a_w 0.45	18	80	Z,M,D,T
Mao1	S17	pelargonic, *Candida* sp., batch 1, 10:3.3	15	96	M,D,T (+c)
Mao2	S18	batch 2	16	96	M,D,T (+c)
Mao3	S19	batch 3	16	96	M,D,T (+c)
Mao4	S20	batch 4 (excluded)	19	120	M,D,T (+c)
Mao5	S21	batch 5 (excluded)	23	144	M,D,T (+c)

empirical basis of the fingerprint idea, that the composition path is a property of the enzyme–substrate pair while the time-scale depends on the conditions, and it is the property that Step 5 tests by transfer. Third, the Mao data lack Z, and in batch 1 the triester reaches a plateau near 90 % instead of 100 %, a feature that mass-conserving models treat, by construction, as a property of the digitised data rather than of the kinetics (Steps 2 and 6).

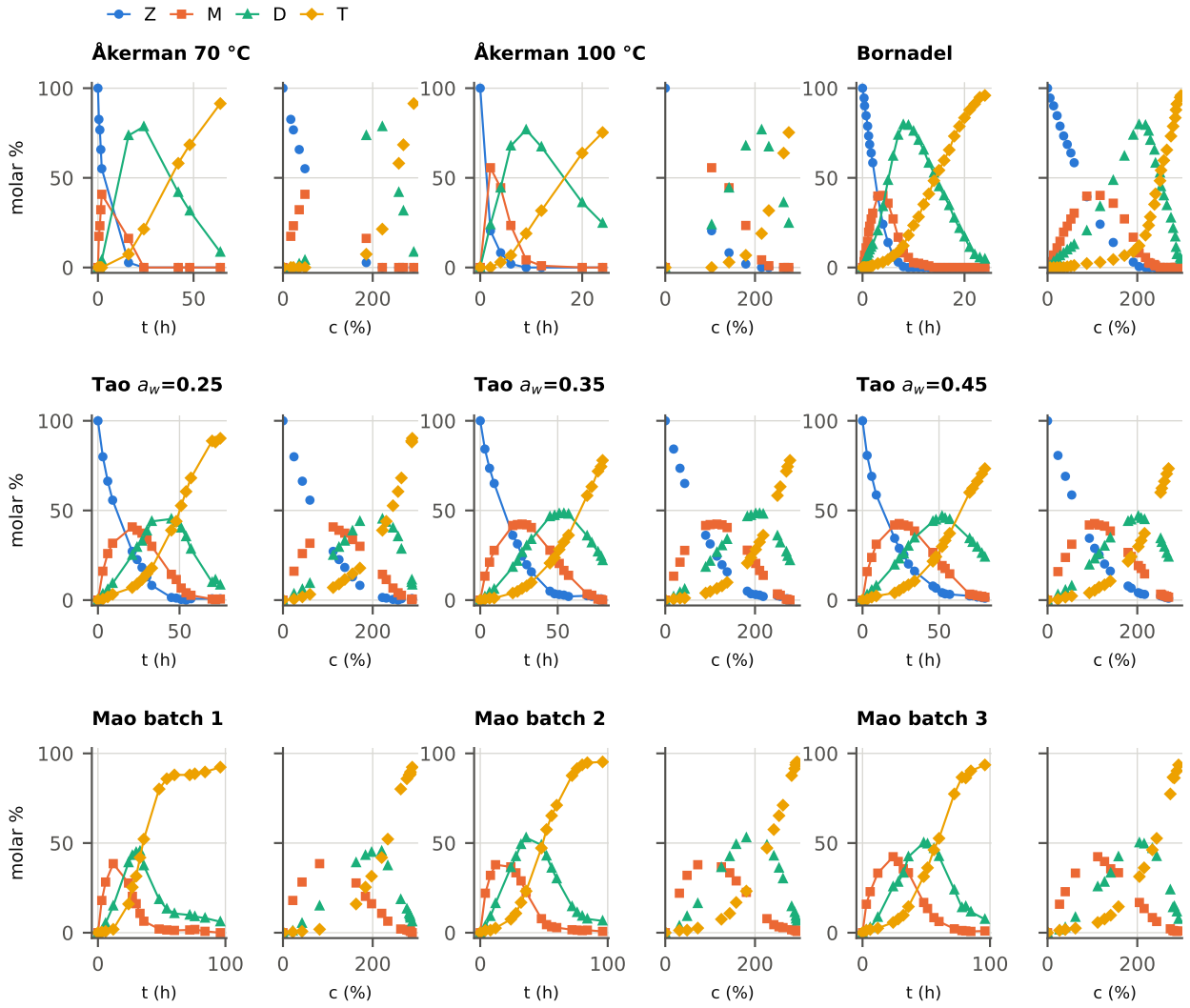


Figure 2: The nine reaction profiles (Supplementary Tables S1–S6, S17–S19 of [10]) against time (left of each pair) and against the degree of reaction c (right). Z, M, D, T: TMP and its mono-, di- and triester (molar % of initial TMP).

Because the paper does not print sample sizes, the public data were checked against every printed quantity that depends on them: the BIC of all twelve dynamic fits, which can only be reproduced with the exact number

of data points n_T, and the sum of squares of the four fingerprint fits, which requires every individual value to match. The printed dynamic parameters were then evaluated in our implementation ("at printed θ"), the same models were re-fitted from 30 multistarts ("global"), 200 random starts of the full model were used to map its local optima, and the profile objective along F_A (all other parameters re-optimised) was computed to test identifiability [19].

2.3 Reference Model: the Original Two-Step Method

For irreversible esterification with water removed, the fingerprinting model of [10] is

$$\begin{aligned} &\frac{dZ}{dc} = \frac{-Z}{S}, \qquad \frac{dM}{dc} = \frac{Z - S_M M}{S}, \qquad \frac{dD}{dc} = \frac{S_M M - S_D D}{S}, \qquad \frac{dT}{dc} = \frac{S_D D}{S}, \\ &S = Z + S_M M + S_D D, \end{aligned} \tag{1}$$

and the dynamic model shares the Ping Pong denominator

$$\begin{aligned} r &= \frac{k_A [E]_T \, A}{A(F_A + F_{AZ} Z + F_{AM} M + F_{AD} D) + Z + S_M M + S_D D}, \\ \frac{dZ}{dt} &= -rZ, \quad \frac{dM}{dt} = r(Z - S_M M), \quad \frac{dD}{dt} = r(S_M M - S_D D), \quad \frac{dT}{dt} = r S_D D, \end{aligned} \tag{2}$$

with $A = A_0 - c$ the fatty acid. Step 1 estimates S_M, S_D by least squares on profiles plotted against c; step 2 fixes them and estimates $k_E = k_A[E]_T$ and the lumped constants against time; the "simplified" model sets the constants that appear unimportant to zero. Because k_A and $[E]_T$ only appear as a product, we estimate k_E and recover the enzyme loading as k_E/k_A; the loadings implied by the printed tables are round numbers (Section 3.1), which confirms that the parameterisation is identical. The objective is $F_{\mathrm{obj}} = \sum (y_{\mathrm{pred}} - y_{\mathrm{obs}})^2$ over Z, M, D, T (Z omitted for Mao) and $\mathrm{BIC} = n_T \ln(F_{\mathrm{obj}}/n_T) + k \ln n_T$ [20]. Following the paper, k counts only the parameters of step 2 ("paper convention"); we additionally report $\mathrm{BIC}_{\mathrm{tot}}$, in which the two selectivities estimated from the same data are also counted. Throughout, "ORIG" denotes the paper's simplified model and "ORIG-full" its full model, both re-fitted by us to their global optimum.

2.4 Generalised Mechanism: Pseudo-Processivity and Deactivation

When the monoester formed in the active site is re-acylated before it diffuses away (pseudo-processivity caused by mass-transfer limitation [13]), one productive encounter can add more than one acyl group. With p_M the probability that a newly formed M is esterified again before release and p_D the analogous probability for D, the event rates per unit of the common factor r become

$$\begin{aligned} \dot{Z} &= -Z, & \dot{M} &= Z(1 - p_M) - S_M M, \\ \dot{D} &= Z p_M (1 - p_D) + S_M M (1 - p_D) - S_D D, & \dot{T} &= Z p_M p_D + S_M M p_D + S_D D, \\ \Sigma \equiv \dot{c} &= Z(1 + p_M + p_M p_D) + S_M M (1 + p_D) + S_D D, \end{aligned} \tag{3}$$

which reduces exactly to Eq. (1) for $p_M = p_D = 0$. Enzyme deactivation inside a batch is described by first-order decay of the active enzyme, $r \to r\, e^{-k_d t}$. Both mechanisms therefore nest the original model: PPK-FIP can only differ from it where the data support p_M, p_D or k_d different from zero, which is what makes them testable.

2.5 Exact Time Reparameterisation

All reactions share the denominator of Eq. (2); therefore the composition is a function of c alone (Eq. 3 divided by Σ), and time follows by quadrature:

$$k_E \, \tau(c) = J_0(c) + F_A J_1(c) + F_{AZ} J_Z(c) + F_{AM} J_M(c) + F_{AD} J_D(c), \qquad \tau = \frac{1 - e^{-k_d t}}{k_d}, \tag{4}$$

with $J_0 = \int_0^c S/((A_0 - c')\Sigma)\, dc'$, $J_1 = \int_0^c dc'/\Sigma$ and $J_X = \int_0^c X/\Sigma \, dc'$ $(X = Z, M, D)$. For $p_M = p_D = 0$, $J_0 = \ln[A_0/(A_0 - c)]$. The four composition equations and five integrals are integrated once on a grid refined towards complete conversion (fourth-order Runge–Kutta compiled with Numba [21]); predictions at any time follow by monotone interpolation of Eq. (4). The rate constants k_E, F_* and k_d then enter only through a scalar product, so changing them costs well below a millisecond.

Figure 3 quantifies what this buys. For one evaluation of the most expensive model (Bornadel design, processive model with deactivation), adaptive ODE solvers need between 19 ms (LSODA at a loose tolerance)

and 1163 ms (Radau at tolerance 10^{-9}); the exact reparameterisation needs 11 ms when the composition path must be recomputed (new S_M, S_D, p_M, p_D) and 0.67 ms when only k_E, F_* or k_d change, at a maximum error of 2×10^{-5} molar % against a Radau reference at tolerance 10^{-12}. The speed-up of one to three orders of magnitude at unchanged accuracy is not a convenience but a precondition of the study: the evaluation programme of Steps 1–6 (30 multistarts per fit, 145 cross-validation folds times every method and ablation variant, and 60 Monte-Carlo replicates per scenario) would otherwise be prohibitive.

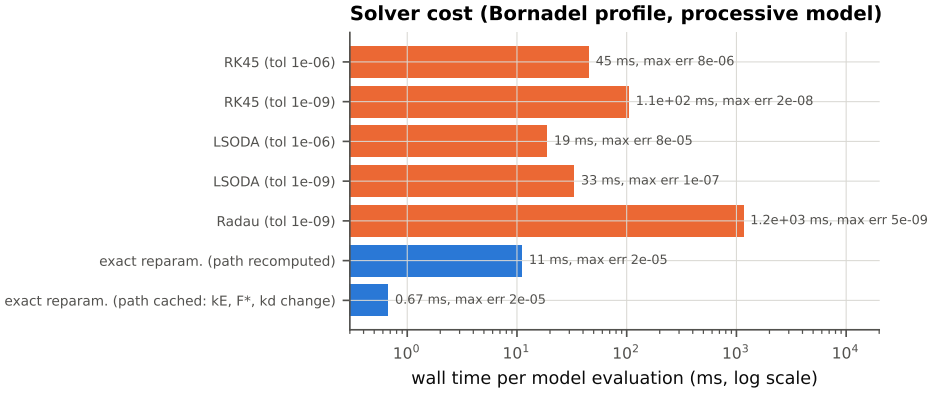


Figure 3: Cost and accuracy of one model evaluation (Bornadel design, processive model with deactivation): adaptive ODE solvers versus the exact reparameterisation, with the composition path recomputed (a new S_M, S_D, p_M, p_D) or cached (new k_E, F_*, k_d). Maximum error relative to Radau at tolerance 10^{-12}.

2.6 Fingerprint-Informed Prior, Model Averaging and Meta-Validation

The two-step method is the limit of a Bayesian estimator in which $(\ln S_M, \ln S_D)$ have an infinitely tight prior at the fingerprint estimate, whereas a joint fit of all parameters corresponds to a flat prior. PPK-FIP uses the intermediate, calibrated choice

$$\ln S_X \sim \mathcal{N}\Big(\ln \hat{S}_X^{\mathrm{fp}},\ (\lambda\, \delta_X/1.96)^2\Big), \quad p_M, p_D \sim \mathcal{N}^+(0, s_p^2), \quad \log_{10} k_d \sim \mathcal{N}(-3, 1.5^2), \tag{5}$$

where δ_X is the relative 95 % half-width of the fingerprint estimate (computed on the training data of each fold), λ widens it, and s_p shrinks processivity towards the original model. The MAP estimate is obtained by appending prior residuals $\hat{\sigma}(u-\mu)/s$ to the least-squares residual vector ($\hat{\sigma}$ = residual SD of the unpenalised fit) and minimising with a trust-region reflective algorithm from warm starts provided by the nested unpenalised family (joint; $+k_d$; +processivity; +both). Two variants are fitted, with the paper's simplified set of lumped constants and with the full set, and their predictions are averaged with BIC weights $w_i \propto e^{-\mathrm{BIC}_i/2}$ [22, 23]. The hyper-parameters $(\lambda, s_p) \in \{1, 3\} \times \{0.05, 0.2\}$ are selected for each case by *leave-one-case-out meta-validation*: the pair that minimises the mean cross-validated error ratio (PPK-FIP/original) over the *other five* case studies is used, so no information from the target case enters the choice.

2.7 Evaluation Protocols

The protocols are listed in the order of the six steps of Table 1. **In-sample (paper protocol; Steps 1–2).** All methods are fitted to the full data of each case with 30 multistarts; we report F_{obj}, BIC (paper convention), $\mathrm{BIC}_{\mathrm{tot}}$ and AICc [24], and examine the residuals and the composition path against c. **Leave-one-time-point-out cross-validation (LOTO; Step 3).** For each of the 145 sampling times, all species measured at that time are removed and every method, including both steps of the original method, is re-estimated from scratch on the remaining data; the held-out values are then predicted. This is the primary out-of-sample criterion. **Forward extrapolation (Step 3).** Each method is trained on the observations with $t \le 0.5\, t_{\max}$ and predicts the rest. With a single split per case this protocol characterises behaviour beyond the data window rather than ranking methods statistically; it is also the protocol that separates mechanistic from data-driven models (Step 6). **Ablation (Step 4).** One component of PPK-FIP at a time is removed (processivity, deactivation, the prior, the averaging, the meta-selection), and every variant is re-estimated in every LOTO fold. A multi-task extension in which profiles of the same system share the composition path (L1), additionally the lumped constants with hierarchical shrinkage (L1h) or everything (L2) is evaluated in the same way, with the level chosen by leave-one-case-out meta-validation. **Cross-condition transfer (Step 5).** The composition path is learned from the other conditions of the same system (other temperature, water activity or batch) and

the held-out profile is predicted against c, with and without processivity. **Monte-Carlo truth recovery (Step 5).** Synthetic data are generated with the sampling design and residual noise of three real cases from three different truths (S1: the original model; S2: pseudo-processive; S3: deactivating), 60 replicates each, and estimation error is measured against the noise-free truth on a dense time grid. **Black-box baselines and hybrid (Step 6).** A Gaussian process (Matérn-5/2 + white noise) [25], a five-member MLP ensemble, the pretrained tabular foundation model TabICL (v2 regressor, in-context learning) [17] and a neural ODE trained through a Runge–Kutta solver [18] receive time, the condition descriptor (water activity or batch) and the known initial state. The hybrid PPK-FIP-GP adds to the mechanistic prediction a Gaussian-process discrepancy of Kennedy–O'Hagan type, fitted for each species to the training residuals, which reverts to zero away from the data; the same discrepancy is added to the original model for comparison.

2.8 Statistics and Software

Paired differences of squared prediction errors are tested with the Wilcoxon signed-rank test [26], Holm-corrected across the six cases [27]; a more conservative test treats each fold rather than each observation as the unit. All computations used Python (NumPy, SciPy [28], scikit-learn, PyTorch) on an NVIDIA DGX Spark (GB10) workstation; all numbers in this paper are generated from the result files by script.

3 Results

3.1 Step 1: The Published Analysis Is Reproducible, but In-Sample Fit Is Not a Reliable Guide

The data are exactly the data of the paper. Table 3 lists every printed quantity that depends on the sample sizes or on individual values. The twelve printed BIC values are recovered exactly when the public numbers of data points n_T (36, 28, 120, 64, 144 and 141) are inserted into the printed F_{obj} and k; a single missing or duplicated row would change them. The four fingerprint sums of squares (74.2, 840.6, 535.3 and 1616.0) are recovered to the printed decimal, which requires every individual value to match. The composition of the data sets (four studies, two temperatures, three water activities, three of five batches used) also agrees. All 36/36 checks passed. Consequently, every comparison in this paper is made on exactly the data the authors used, and any difference between methods is due to the method.

Step 1 of the original method is reproduced exactly. Table 4 compares the printed and reproduced fingerprint estimates: all four pairs of selectivities and all four sums of squares agree to the printed precision, and the 95 % confidence intervals agree within 2–12 %, a difference attributable to the finite-difference Hessian used by the authors. Figure 4a shows the same agreement graphically: the printed and reproduced values of S_M, S_D and SS (circles) and of the simplified dynamic fits of Åkerman and Bornadel (squares) lie on the identity line over six orders of magnitude. The fingerprint estimates that serve as prior centres in PPK-FIP are therefore the authors' own numbers.

Step 2 is reproduced where the printed fits are at their optimum, and reveals two problems where they are not. Table 5 gives, for each of the twelve dynamic fits, the printed F_{obj}, its value when the printed parameters are inserted into our implementation, the implied enzyme loading $E = k_E/k_A$, and our global optimum. The simplified models of Åkerman et al. and the Bornadel models are reproduced to the last digit (F_{obj}, BIC, F_{AZ}, F_{AM}, F_{AD}), and evaluating the model at the printed parameters recovers the printed F_{obj} in 7 of 12 tables. The implied loadings are round numbers, 2.00 % for Åkerman, 5.00 % for Bornadel and 15.0 % for Mao (Figure 4c, where the points fall on the grid lines at 2, 5 and 15), which confirms that the parameterisation is identical; for the Mao case the printed full model is recovered ($F_{\mathrm{obj}} = 1476.5$ vs. 1475.9) only with the initial acid:TMP ratio of the actual charge, 10:3.3 ($A_0 = 303.0$), not with the nominal 3:1. Two discrepancies remain, and both are visible in Figure 4b, which shows for every fit the printed objective (orange), the objective at the printed parameters (yellow) and our global optimum (blue). (i) Several printed full models are not optimal: the blue bar lies below the orange one for the Åkerman 100 °C, Tao and Mao full models, most strikingly for Åkerman 100 °C (48.9 vs. 117.9, a factor of 2.4). (ii) For the Tao case the yellow bars rise far above the orange ones: the printed dynamic fits (121.2/123.5) could not be reproduced from the published profiles, the best value at the printed parameters over all initial acid ratios being 131.1.

Figure 5 explains why the full model is prone to non-optimal fits. Panel (a) shows the local optima reached from 200 random starts of the original full model for each case, as $\log_{10}$ of the ratio to the global minimum, together with the value printed in the paper (diamond). Only 46–94 % of the starts reach the global minimum, the rest end on plateaus up to three orders of magnitude higher, and the printed Åkerman 100 °C value sits on one of the secondary optima ($\log_{10}$ ratio 0.38). Panel (b) shows the profile objective along F_A with all other parameters re-optimised: for the three cases shown it decreases monotonically towards $F_A \to \infty$ and never turns up. The full model is therefore structurally non-identifiable [19, 29]: in the limit $F_A \to \infty$ the rate

Table 3: Hard data check: every quantity printed in the paper that depends on the sample sizes or on individual values.

Check	Paper	Public data	Match
Table 1 Akerman70 full: printed BIC recomputed with public nT=36 (Fobj=59.0, k=5)	35.7	35.7	YES
Table 1 Akerman100 full: printed BIC recomputed with public nT=28 (Fobj=117.9, k=5)	56.9	56.9	YES
Table 2 Akerman70 simplified: printed BIC recomputed with public nT=36 (Fobj=70.9, k=3)	35.1	35.1	YES
Table 2 Akerman100 simplified: printed BIC recomputed with public nT=28 (Fobj=118.0, k=3)	50.3	50.3	YES
Table 3 Bornadel full: printed BIC recomputed with public nT=120 (Fobj=329.9, k=5)	145.3	145.3	YES
Table 3 Bornadel simplified: printed BIC recomputed with public nT=120 (Fobj=329.9, k=4)	140.5	140.5	YES
Table 4 Tao025 full: printed BIC recomputed with public nT=64 (Fobj=121.2, k=5)	61.7	61.7	YES
Table 4 Tao035+045 full: printed BIC recomputed with public nT=144 (Fobj=325.8, k=5)	142.4	142.4	YES
Table 5 Tao025 simplified: printed BIC recomputed with public nT=64 (Fobj=123.5, k=3)	54.5	54.5	YES
Table 5 Tao035+045 simplified: printed BIC recomputed with public nT=144 (Fobj=326.7, k=3)	132.9	132.9	YES
Table 6 Mao1-3 full: printed BIC recomputed with public nT=141 (Fobj=1475.9, k=7)	365.7	365.7	YES
Table 6 Mao1-3 simplified: printed BIC recomputed with public nT=141 (Fobj=1771.5, k=5)	381.6	381.6	YES
Number of literature sources / case studies	4	4	YES
Akerman datasets (70 and 100 C)	2	2	YES
Tao datasets (aw 0.25/0.35/0.45)	3	3	YES
Mao batches tabulated / used	5 / 3	5 / 3	YES
Fingerprinting fit Akerman: sum of squares reproduced on public data	74.2	74.2	YES
Fingerprinting fit Bornadel: sum of squares reproduced on public data	840.6	840.6	YES
Fingerprinting fit Tao: sum of squares reproduced on public data	535.3	535.3	YES
Fingerprinting fit Mao: sum of squares reproduced on public data	1616.0	1616.0	YES

Table 4: Step 1 (fingerprinting) reproduced: printed / reproduced.

System	S_M	±95% CI	S_D	±95% CI	SS
Åkerman	0.389 / 0.389	0.026 / 0.025	0.0311 / 0.0311	0.0035 / 0.0035	74.2 / 74.2
Bornadel	0.909 / 0.909	0.096 / 0.094	0.0660 / 0.0657	0.0110 / 0.0122	840.6 / 840.6
Tao	0.761 / 0.761	0.025 / 0.024	0.3330 / 0.3330	0.0120 / 0.0113	535.3 / 535.3
Mao	0.916 / 0.916	0.094 / 0.085	0.3780 / 0.3783	0.0370 / 0.0326	1616.0 / 1616.0

Table 5: Step 2 (dynamic model) audit. "At printed θ": objective evaluated at the printed parameters with the enzyme loading $E = k_E/k_A$ optimised (one scalar per case); "global": our 30-start optimum of the same model; "consistent": printed F_{obj} recovered within 1.5% at the printed parameters.

Case	Model	F_{obj} printed	at printed θ	E	F_{obj} global	BIC printed	BIC global	consistent
Åkerman 70 °C	full	59.0	59.0	2.00	58.4	35.7	35.3	yes
Åkerman 100 °C	full	117.9	118.0	2.00	48.9	56.9	32.3	yes
Åkerman 70 °C	simplified	70.9	70.9	2.00	70.9	35.1	35.1	yes
Åkerman 100 °C	simplified	118.0	118.0	2.01	118.0	50.3	50.3	yes
Bornadel	full	329.9	329.9	5.00	329.9	145.3	145.3	yes
Bornadel	simplified	329.9	329.9	5.00	329.9	140.5	140.5	yes
Tao a_w=0.25	full	121.2	458.2	0.09	110.8	61.7	55.9	no
Tao a_w=0.35+0.45	full	325.8	423.8	0.09	319.6	142.4	139.7	no
Tao a_w=0.25	simplified	123.5	590.5	0.09	202.1	54.5	86.1	no
Tao a_w=0.35+0.45	simplified	326.7	448.8	0.09	337.9	132.9	137.7	no
Mao batches 1–3	full	1475.9	1476.5	15.02	1373.2	365.7	355.6	yes
Mao batches 1–3	simplified	1771.5	1631.4	15.03	1624.4	381.6	369.4	no

becomes independent of acid saturation and k_E and the F constants can only be estimated as ratios, which is also why the printed confidence intervals of the full models are so large and why our own full-set fits sit at the parameter bound (Table 7 below).

Role in the argument. Two conclusions carry forward. First, the data and the reference implementation are exact, so every later difference between methods is attributable to the estimator. Second, the model-selection evidence of the original paper rests on in-sample statistics evaluated partly at non-optimal points of a non-identifiable model, and one case cannot be reproduced at all. This motivates three decisions: all comparisons

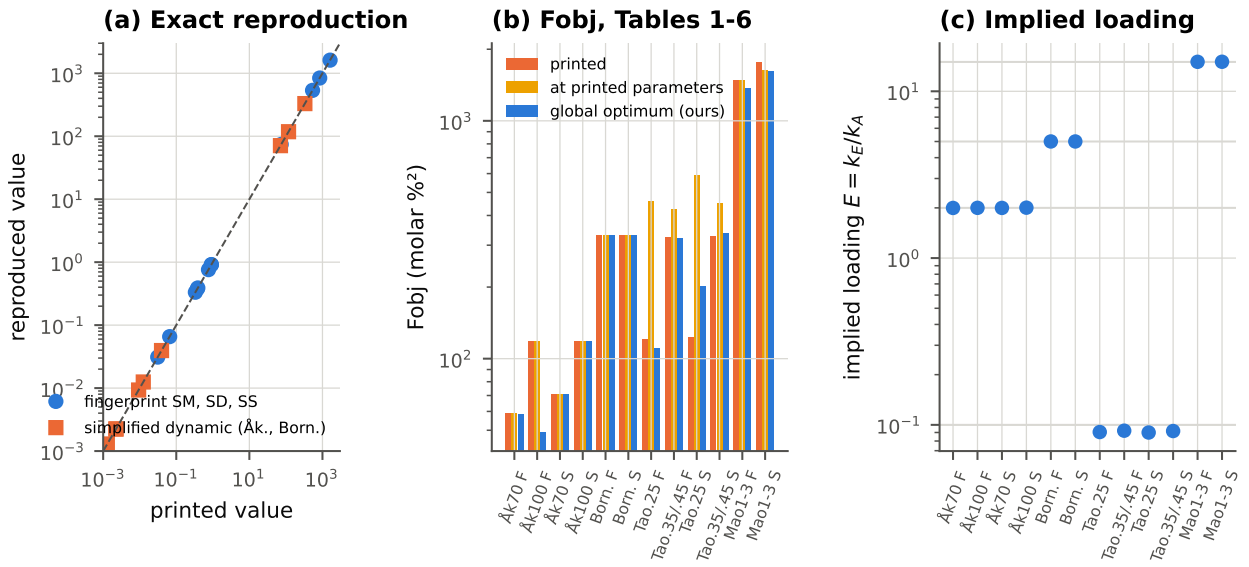


Figure 4: Reproduction audit. (a) Printed versus reproduced fingerprint and simplified-model quantities (log–log). (b) F_{obj} printed, evaluated at the printed parameters, and at our global optimum (F: full model, S: simplified model). (c) Enzyme loading implied by k_E/k_A at the printed parameters; grid lines at 0.1, 2, 5 and 15.

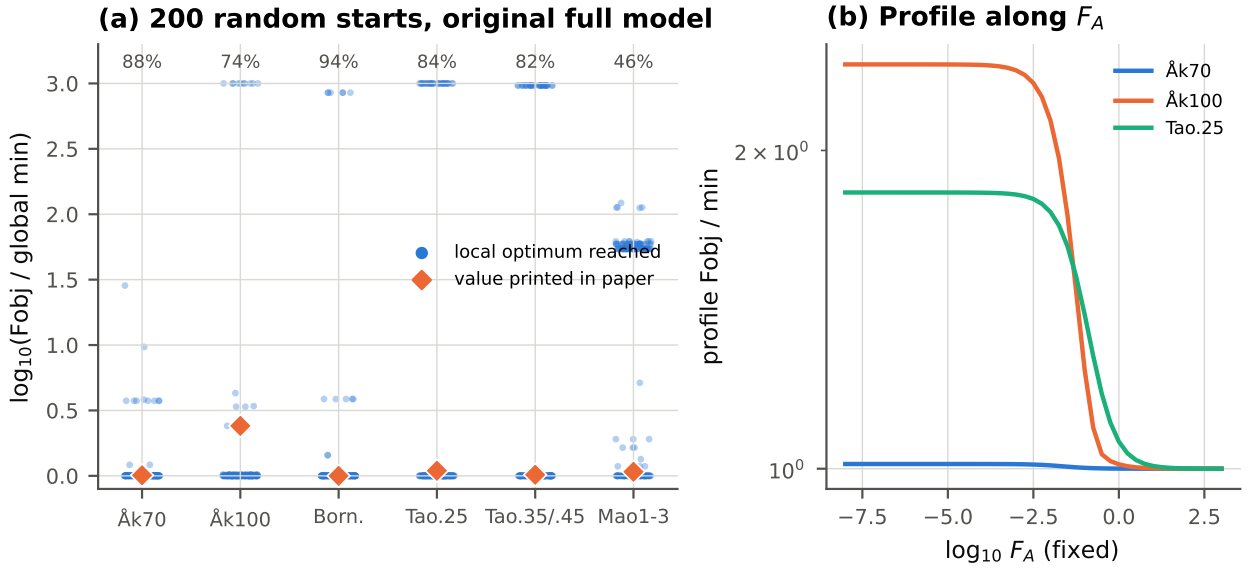


Figure 5: (a) Local optima reached from 200 random starts of the original full dynamic model (dots), with the value printed in the paper (diamonds); numbers above give the percentage of starts that reach the global minimum. (b) Profile objective along F_A (other parameters re-optimised), relative to its minimum: the objective decreases monotonically towards $F_A \to \infty$, i.e. the full model is not identifiable.

Table 6: In-sample fit on the full data (paper protocol). Paper columns: printed F_{obj} / printed BIC (paper convention, selectivities not counted). Refit and PPK-FIP columns: F_{obj} / BIC_{tot} counting every estimated parameter (for PPK-FIP the BIC-weighted mean parameter count $\bar{k}$).

Case	Paper simpl. F_{obj}/BIC	Paper full	ORIG refit F_{obj}/BIC_{tot}	ORIG-full refit	PPK-FIP	$\bar{k}$
Åkerman 70 °C	70.9 / 35.1	59.0 / 35.7	70.9 / 42.3	58.4 / 42.5	**56.8** / **48.8**	9.0
Åkerman 100 °C	118.0 / 50.3	117.9 / 56.9	118.0 / 56.9	48.9 / 38.9	**31.8** / **33.2**	8.9
Bornadel	329.9 / 140.5	329.9 / 145.3	329.9 / 150.1	329.9 / 154.9	**105.0** / **27.5**	9.1
Tao a_w=0.25	123.5 / 54.5	121.2 / 61.7	202.1 / 94.4	110.8 / 64.2	**75.1** / **51.7**	10.0
Tao a_w=0.35+0.45	326.7 / 132.9	325.8 / 142.4	337.9 / 147.7	319.6 / 149.6	**260.2** / **133.8**	9.8
Mao batches 1–3	1771.5 / 381.6	1475.9 / 365.7	1624.4 / 379.3	1373.2 / 365.5	**1169.6** / **357.7**	12.0

use our global re-fits ORIG and ORIG-full as references, which are at least as good as the printed fit of the same model in every case; BIC is also reported with all estimated parameters counted; and the decisive criteria are out-of-sample (Step 3).

3.2 Step 2: The Remaining Misfit Has a Mechanistic Signature

Table 6 and Figure 6 compare the fits to the full data, the protocol of the original paper. Figure 6a normalises each F_{obj} by the best value printed in the paper for that case: the ORIG refit (orange) equals or exceeds 1 because the simplified form is more restrictive than the best printed form, the ORIG-full refit (yellow) is at or below 1 because it reaches the global optimum, and PPK-FIP (blue) is lowest in every case, at 0.96, 0.27, 0.32, 0.62, 0.80 and 0.79 of the printed value. Figure 6b shows BIC_{tot} relative to the re-fitted ORIG model, with all estimated parameters counted: PPK-FIP is lower in 5 of 6 cases, by 14 to 123 units, and differs only for Åkerman 70 °C (+6), where the penalty for the two mechanisms that the data set to zero is counted. The gains are large but unevenly distributed, and the rest of this section locates them.

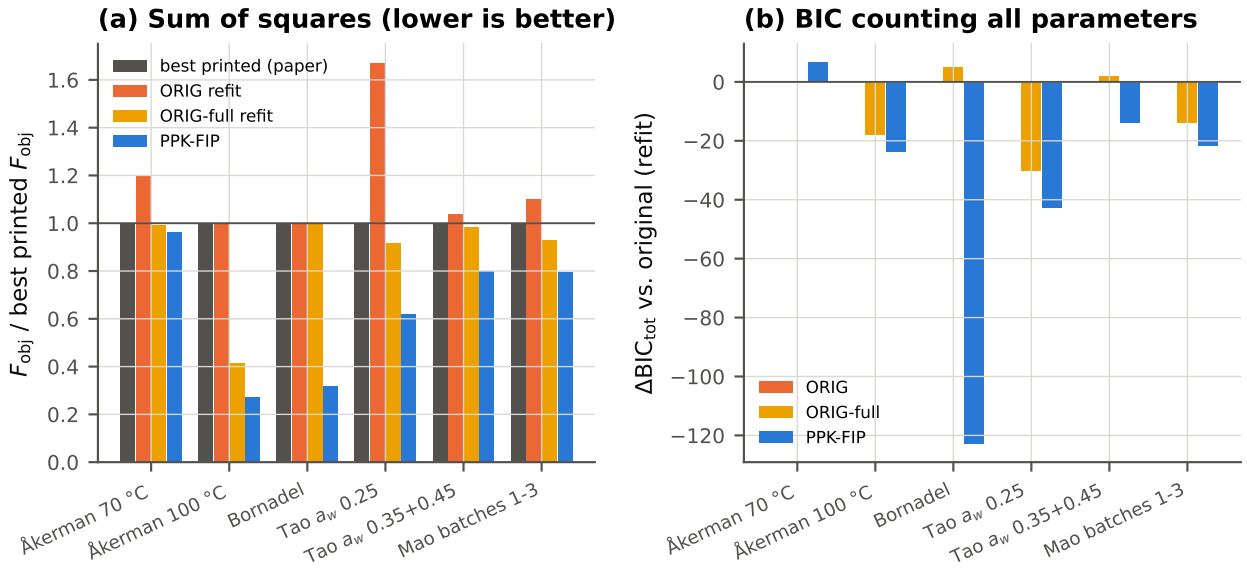


Figure 6: In-sample comparison. (a) F_{obj} relative to the best value printed in the paper. (b) BIC_{tot} (all estimated parameters counted) relative to the re-fitted original simplified model; negative is better.

Figure 7 shows the fitted curves of the re-fitted original method (dashed) and PPK-FIP (solid) for all nine profiles, Figure 8 the composition paths against c for the three profiles in which the new mechanisms are active, Figure 9 the residuals, and Table 7 the estimates.

Åkerman et al. At 70 °C all models describe the nine sampling times well (F_{obj} 56.8 vs. 59.0 printed) and the two curves in Figure 7 are indistinguishable; PPK-FIP sets processivity to zero ($p_M = 0$) and keeps deactivation at the prior level ($k_d = 6.3e-04\,h^{-1}$), so that its only difference from the original model is the BIC penalty for two mechanisms it correctly leaves inactive, a first illustration that PPK-FIP does not add mechanisms where the data do not call for them. At 100 °C the improvement is large, from 117.9 to 31.8 (73.0 %), and it comes from the second half of the profile, where the reaction slows down more than acid depletion alone can explain. PPK-FIP attributes this to first-order deactivation with $k_d = 0.018\,h^{-1}$ (BIC-

Table 7: PPK-FIP estimates on the full data for the dominant model form (F set with the larger BIC weight; w_{simpl} = weight of the simplified form) with linearised 95% confidence half-widths computed as in the original paper. Values of k_E at the upper bound (1000) indicate the non-identifiable ridge of the full F set (Figure 5b), along which only ratios of k_E and the F constants are determined.

Profile	S_M	S_D	p_M	p_D	k_d (h^{-1})	k_E	F set	w_{simpl}
Åkerman 70	0.393 ± 0.197	0.0264 ± 0.0268	0.000 ± 0.0526	0.016 ± 0.0728	1.08e-03	1e+03	f	0.48
Åkerman 100	0.385 ± 0.0525	0.0291 ± 0.0102	0.000 ± 0.0426	0.040 ± 0.0354	3.16e-02	0.394	s	0.56
Bornadel	0.563 ± 0.0493	0.0394 ± 0.00713	0.175 ± 0.0245	0.013 ± 0.0182	2.21e-04	0.993	s	0.92
Tao025	0.760 ± 0.0722	0.2981 ± 0.0383	0.030 ± 0.0496	0.062 ± 0.0361	3.15e-04	302	f	0.02
Tao035	0.731 ± 0.0551	0.2810 ± 0.0293	0.007 ± 0.0383	0.053 ± 0.0292	2.15e-04	227	f	0.11
Tao045	0.731 ± 0.0551	0.2810 ± 0.0293	0.007 ± 0.0383	0.053 ± 0.0292	2.15e-04	227	f	0.11
Mao1	0.829 ± 0.157	0.2932 ± 0.0728	0.000 ± 0.0592	0.028 ± 0.0665	3.14e-04	297	f	0.00
Mao2	0.829 ± 0.157	0.2932 ± 0.0728	0.000 ± 0.0592	0.028 ± 0.0665	3.14e-04	220	f	0.00
Mao3	0.829 ± 0.157	0.2932 ± 0.0728	0.000 ± 0.0592	0.028 ± 0.0665	3.14e-04	188	f	0.00

weighted over the two model forms; $0.032\,\mathrm{h}^{-1}$ in the dominant simplified form, Table 7), i.e. a half-life of the active enzyme of roughly one day at 100 °C, against no detectable loss at 70 °C.

Bornadel et al. This is the case the original authors flagged as mass-transfer limited, and the data select pseudo-processivity decisively: $p_M = 0.18$ with a 95 % half-width of 0.025 (Table 7), $p_D = 0.013$, and F_{obj} falls from 329.9 to 105.0 (68.2 %) while $\mathrm{BIC}_{\mathrm{tot}}$ falls by more than 120 units. Figure 9b shows the mechanism at work in the residuals of M and D: the original model (circles, dashed) over-predicts M by 2–5 molar % and under-predicts D by 4–5 molar % during the first two hours, then reverses sign around 5 h, a systematic pattern that no choice of lumped constants can remove because it is fixed by the composition path; with pseudo-processivity (squares, solid) the early residuals shrink to within ±2 molar % and lose their systematic sign. Figure 8 (left) shows the same thing in the c-domain: the fingerprint path of the original method (dashed) rises too early in M and too late in D, whereas the processive path (solid) follows the data. The selectivities change accordingly, from the apparent values $S_M = 0.909$, $S_D = 0.066$ to $S_M = 0.56$, $S_D = 0.039$, in agreement with the processive fingerprint analysis of Mitchell et al. [13]; in Figure 7 the two curves look alike at full scale because the difference is concentrated in the first eight hours and is spread over 30 sampling times.

Tao et al. At $a_w = 0.25$, F_{obj} decreases from 121.2 (printed) to 75.1 (38.0 %), mainly through the full set of lumped constants (weight of the simplified form 0.02, Table 7) and a small processivity of the diester step ($p_D = 0.06$); the middle panel of Figure 8 shows a correspondingly small shift of the path. At $a_w = 0.35/0.45$ the decrease is from 325.8 to 260.2 (20.1 %).

Mao et al. The three reuse batches are fitted with batch-specific k_E as in the original paper; F_{obj} decreases from 1475.9 to 1169.6 (20.8 %). The remaining misfit is dominated by the late T and D values of batch 1 (Figure 7, bottom left): the measured triester plateaus near 90 % while D stays at 5–10 % after 70 h, whereas any model that conserves TMP and lets the reaction run to completion must send T to 100 % and D to zero; the residual that remains there is therefore a property of the digitised data rather than of the kinetics, as anticipated in Section 2, and the same feature explains the interpolation results of Step 6. Figure 9a summarises the residuals of all six cases: the median absolute residual of PPK-FIP is below that of the original model in every case, the largest reductions of the upper quartile are for Åkerman 100 °C (2.7 to 1.1 molar %) and Bornadel (1.7 to 1.0), and the Mao whiskers, which are set by the batch-1 plateau, shorten from 7.7 to 5.9.

Table 7 makes the pattern explicit across all profiles. The processivity p_M is zero within its confidence half-width everywhere except Bornadel, where it is seven standard errors from zero; p_D is small (0.01–0.06) and clearly non-zero only for Tao $a_w = 0.25$ and Åkerman 100 °C; k_d stays at the prior level (10^{-4}–$10^{-3}\,\mathrm{h}^{-1}$) everywhere except Åkerman 100 °C; the selectivities move away from the fingerprint values only where processivity is present; and the full lumped-constant set, where it dominates, sits on the identifiability ridge (k_E at its bound), confirming Figure 5b.

Role in the argument. The in-sample gains are large, but more importantly they are localised and interpretable. One mechanism, pseudo-processivity, acts in the mass-transfer-limited Bornadel data, exactly where the original authors anticipated it, and removes a residual pattern that lumped constants cannot touch; the other, deactivation, acts in the single profile that slows down late; and almost nothing changes where the original model was adequate (Åkerman 70 °C). PPK-FIP therefore does not fit better everywhere through added flexibility; it fits better where a specific mechanism is missing. Whether this translates into prediction is the question of Step 3.

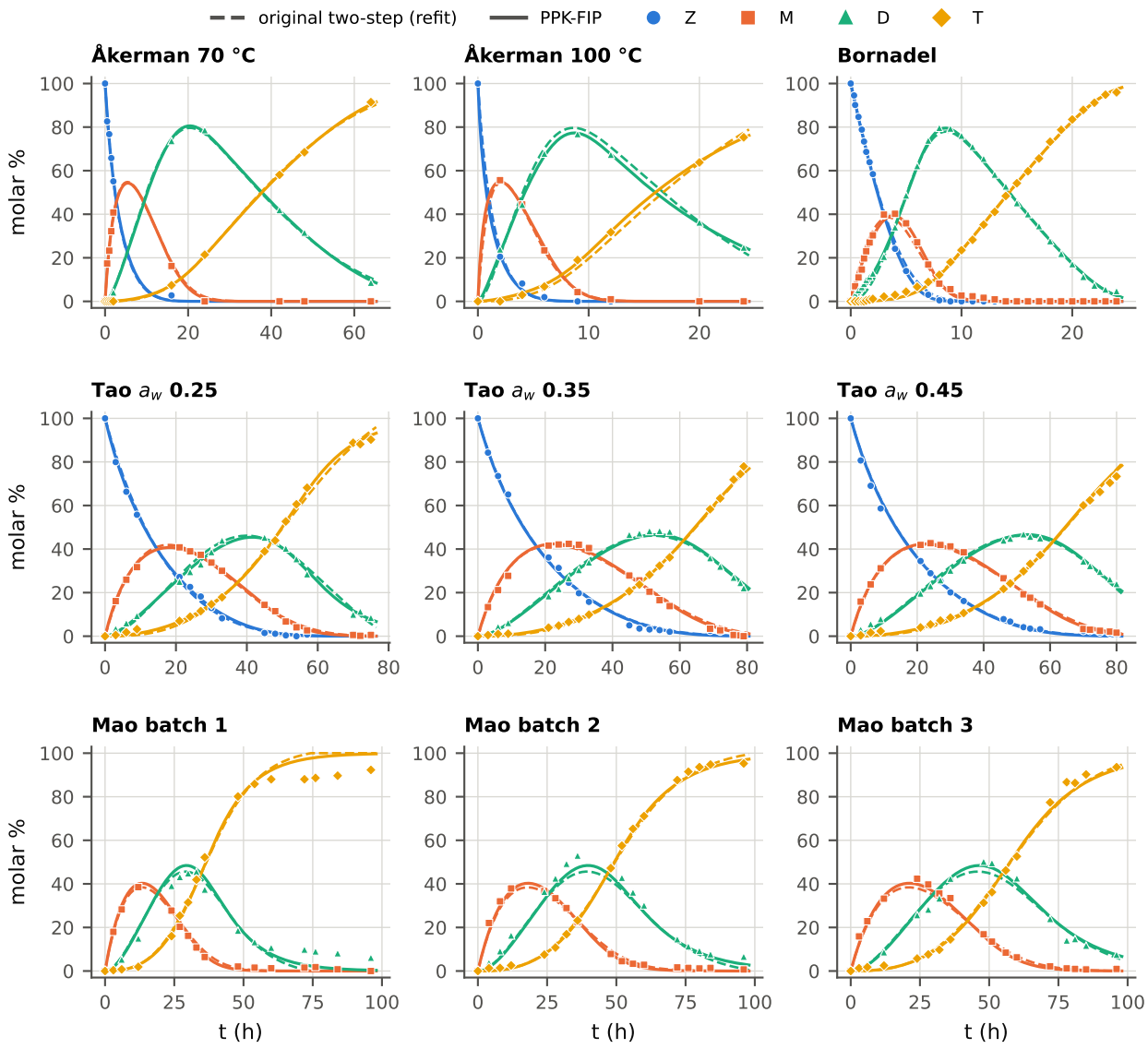


Figure 7: Fits to the full data: original two-step method re-fitted to its global optimum (dashed) and PPK-FIP (solid). Markers: data.

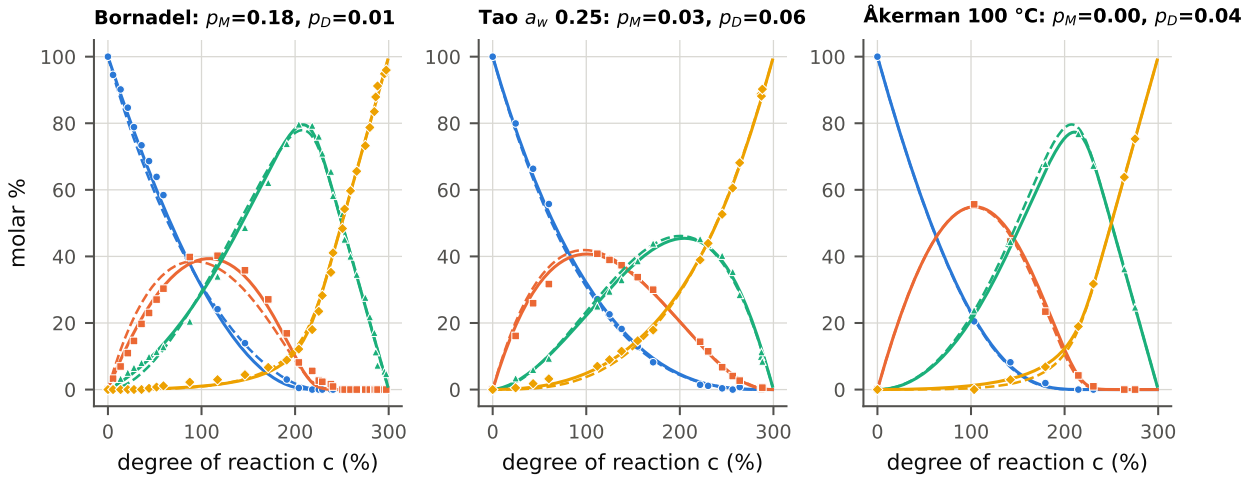


Figure 8: Composition paths against the degree of reaction: fingerprint path of the original method (dashed) and PPK-FIP path with pseudo-processivity (solid); estimated processivities in the panel titles. Colours as in Figure 7.

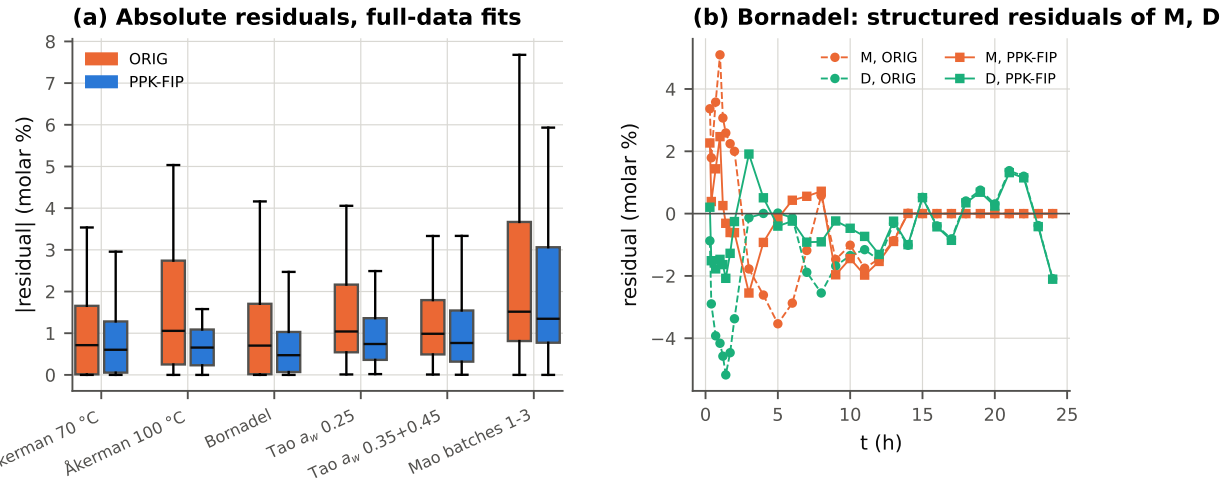


Figure 9: (a) Absolute residuals of the full-data fits. (b) Residuals of M and D for the Bornadel profile: the systematic early pattern of the original model (circles, dashed) is removed by pseudo-processivity (squares, solid).

Table 8: Leave-one-time-point-out cross-validation: RMSE (molar %) of every procedure re-estimated from scratch in each fold. ratio = RMSE(PPK-FIP)/RMSE(reference); p = Wilcoxon signed-rank test on paired squared errors, Holm-corrected over the six cases (O: vs. original simplified, F: vs. original full model; pooled row uncorrected).

Case	ORIG	ORIG-full	ORIG-BIC	JOINT-BMA	PPK-FIP	ratio$_O$	p_O	ratio$_F$	p_F
Åkerman 70 °C	1.828	**1.655**	1.875	2.342	1.886	1.032	0.568	1.140	0.035
Åkerman 100 °C	3.742	**3.293**	**3.293**	3.882	3.505	0.937	0.568	1.065	0.522
Bornadel	1.825	1.825	1.825	**1.052**	1.054	0.578	<0.001	0.578	<0.001
Tao a_w=0.25	2.101	1.440	1.440	**1.310**	1.358	0.646	<0.001	0.943	0.232
Tao a_w=0.35+0.45	1.643	1.591	1.659	**1.459**	1.495	0.910	<0.001	0.940	0.008
Mao batches 1–3	3.607	3.314	3.314	3.104	**3.098**	0.859	<0.001	0.935	0.015
All cases (pooled)	2.533	2.312	2.336	2.177	**2.121**	0.838	<0.001	0.918	<0.001

3.3 Step 3: The Gain Survives Out-of-Sample

Lower in-sample error is expected for a richer model; the decisive question is whether it predicts data not used for fitting. Table 8 and Figure 10 report the leave-one-time-point-out cross-validation, in which both steps of the original method are re-estimated in every one of the 145 folds. Figure 10a shows the RMSE per case for the original simplified and full models, the joint fit without prior (JOINT-BMA) and PPK-FIP; Figure 10b shows the ratio of PPK-FIP to each original form on a log axis, with Holm-corrected significance stars coloured by the direction of the paired median. Pooled over all 533 held-out observations, PPK-FIP reduced the RMSE from 2.533 to 2.121 molar % (16.2 %, p <0.001) relative to the original simplified method and from 2.312 by 8.2 % (p <0.001) relative to the original full model. Per case, PPK-FIP had a lower RMSE than the original method in 5 of 6 cases (4 significant after Holm correction, none significantly worse) and lower than the full model in 4 of 6 (3 significantly better). The largest gains are for Bornadel (1.825→1.054), Tao a_w 0.25 (2.101→1.358) and Mao (3.607→3.098), i.e. the cases in which Step 2 located a missing mechanism or a dominant full F set; for Åkerman 70 °C, the profile with only nine sampling times, the three estimators lie within 0.25 molar % of one another (1.886, 1.828 and 1.655). Figure 10a also previews Step 4: the joint fit without prior is as good as PPK-FIP on the densely sampled profiles but clearly worse on the two sparse Åkerman profiles (2.34 vs. 1.89 and 3.88 vs. 3.51). A more conservative test that treats each fold rather than each observation as the unit confirms the result: PPK-FIP has the lower fold error in 108 of 145 folds against the original method ($p = 2.6 \times 10^{-10}$) and in 97 of 145 against the full model ($p = 9.4 \times 10^{-6}$).

Forward extrapolation probes behaviour beyond the training window (Table 9, Figure 11): trained on the first half of each profile, every method predicts the second half with errors of 1–20 molar % instead of the 1–4 of interpolation. PPK-FIP halved the extrapolation error for Åkerman 100 °C (10.670→5.506), where the deactivation identified in Step 2 is already partly expressed within the training window, reduced it for Tao a_w 0.25 (19.861→17.525) and for Bornadel and Tao a_w 0.35/0.45, and pooled over all cases the two mechanistic estimators are equivalent (11.874 vs. 12.080, $p = 0.184$). For Åkerman 70 °C (1.097 vs. 1.443) and the Mao batches (4.654 vs. 7.765) the ranking reverses, and the reason is informative: the first half of a short profile contains the rise of M and D but not the late behaviour from which k_d and the processivities are identified, so

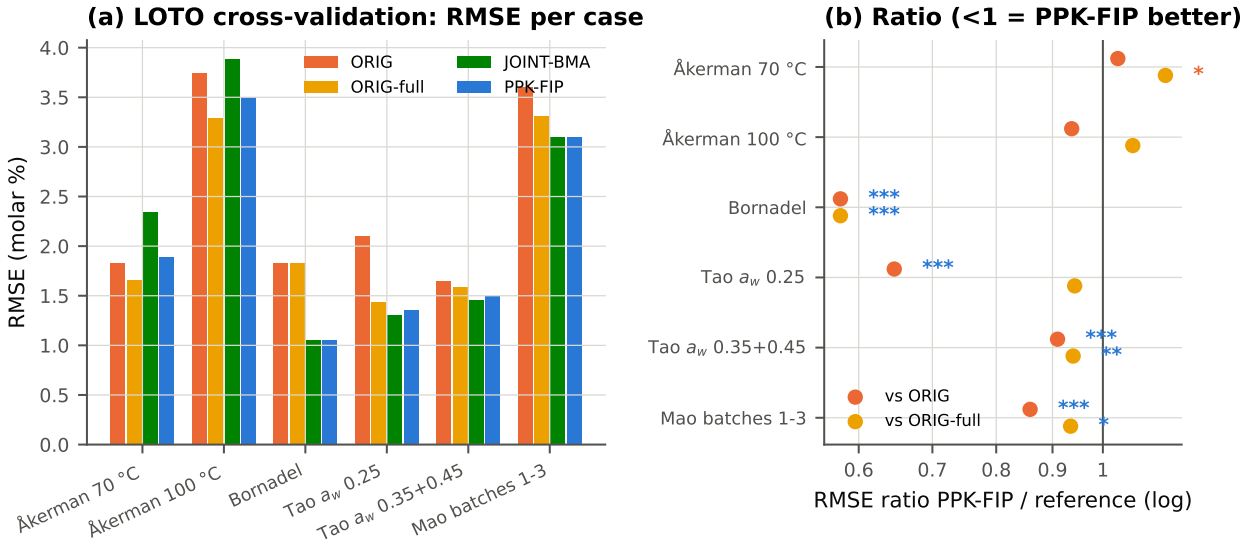


Figure 10: Leave-one-time-point-out cross-validation. (a) RMSE per case. (b) RMSE ratio PPK-FIP/reference; stars mark Holm-corrected Wilcoxon tests (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$), blue when the paired median favours PPK-FIP and orange otherwise.

Table 9: Forward extrapolation (train $t \leq 0.5 t_{\max}$): RMSE (molar %) of every procedure re-estimated from scratch in each fold. ratio = RMSE(PPK-FIP)/RMSE(reference); p = Wilcoxon signed-rank test on paired squared errors, Holm-corrected over the six cases (O: vs. original simplified, F: vs. original full model; pooled row uncorrected).

Case	ORIG	ORIG-full	ORIG-BIC	JOINT-BMA	PPK-FIP	ratio$_O$	p_O	ratio$_F$	p_F
Åkerman 70 °C	**1.097**	9.527	**1.097**	1.733	1.443	1.315	1.000	0.152	0.003
Åkerman 100 °C	10.670	**0.673**	10.670	5.773	5.506	0.516	1.000	8.180	0.594
Bornadel	4.044	4.044	4.044	4.415	**3.971**	0.982	1.000	0.982	1.000
Tao a_w=0.25	19.861	19.861	19.861	**17.167**	17.525	0.882	1.000	0.882	1.000
Tao a_w=0.35+0.45	15.866	15.866	15.866	15.897	**15.753**	0.993	1.000	0.993	0.897
Mao batches 1–3	4.654	**4.159**	4.654	5.660	7.765	1.669	1.000	1.867	0.013
All cases (pooled)	12.080	12.060	12.080	**11.582**	11.874	0.983	0.184	0.985	0.740

the protocol defines the data requirement of the additional mechanisms, namely a profile that extends into the slow-down, as the ablation of Step 4 confirms. The original full model illustrates the opposite problem, with results ranging from 0.67 (Åkerman 100 °C) to 9.5 (Åkerman 70 °C), as expected of a non-identifiable model whose ridge is pinned down differently by different halves of the data. Figure 12 shows three representative profiles with the training window shaded. For Åkerman 100 °C PPK-FIP (solid) comes closest to the measured T at 24 h among the mechanistic models; for Bornadel both mechanistic models track T and D closely, whereas the neural ODE (dotted) lets M rise again after 15 h, which is physically impossible and reflects the absence of any conservation constraint; for Tao a_w 0.25 the first 38 h of data contain no information about the later slow-down, so both mechanistic models place complete conversion earlier than observed and the neural ODE extrapolates in the opposite direction, which marks the information limit of a half-profile.

Role in the argument. Leave-one-time-point-out validation is the decisive test of the paper, and PPK-FIP passes it: a 16.2 % lower prediction error than the original method, significant in four cases and never significantly worse, with the gain concentrated where Step 2 found a missing mechanism. Forward extrapolation, with one split per case, places the two mechanistic estimators on a par and adds two pieces of information: the additional mechanisms need a profile that extends into the slow-down to be identified, and a half-profile without that information (Tao a_w 0.25) bounds every estimator alike. It is also the protocol that separates mechanistic from black-box models in Step 6.

3.4 Step 4: Processivity and the Prior Carry the Gain; Deactivation and Averaging Do Not

Table 10 and Figure 13 remove one component of PPK-FIP at a time, re-estimating every variant in every LOTO fold. Figure 13a shows the pooled LOTO RMSE of each variant against the dashed line of full PPK-

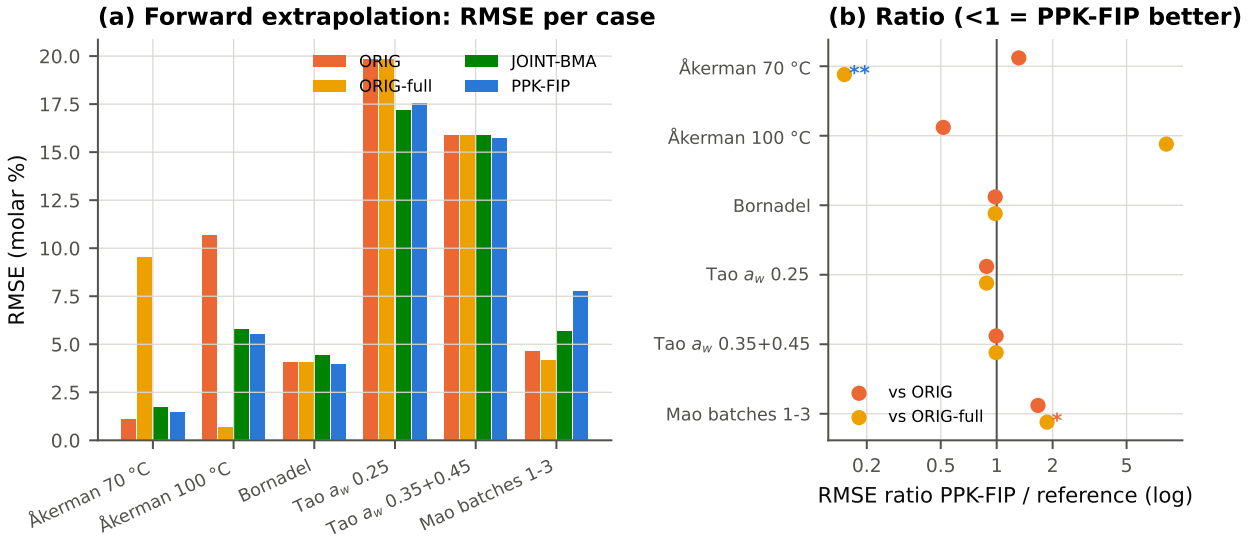


Figure 11: Forward extrapolation (training on $t \leq 0.5\,t_{\max}$); layout as in Figure 10.

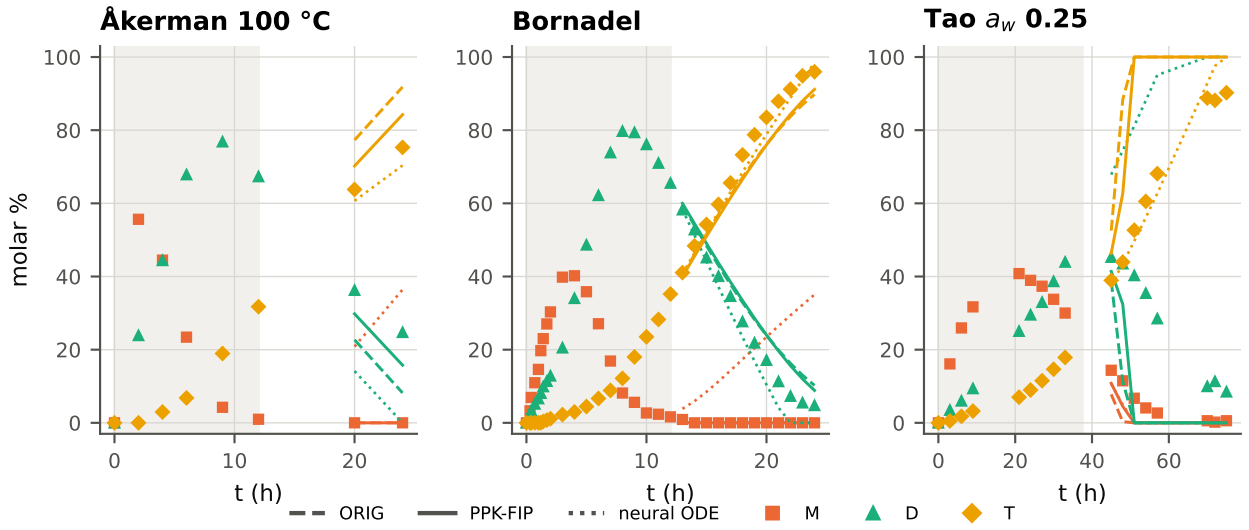


Figure 12: Forward extrapolation for three profiles; the shaded region was used for training. Lines connect the predictions at the held-out sampling times of the original method (dashed), PPK-FIP (solid) and the black-box neural ODE (dotted).

FIP (2.121), with the two original two-step forms in orange; Figure 13b shows the per-case ratio of each variant to PPK-FIP as a heat map, red where the variant is worse. Four results follow. (1) Removing pseudo-processivity raises the pooled RMSE to 2.289 ($p<0.001$), and the heat map shows that the loss is almost entirely the Bornadel column (1.718 vs. 1.054, ratio 1.63), with small losses for Åkerman 100 °C and Tao; this mechanism carries most of the out-of-sample gain. (2) Within-batch deactivation is specific to the Åkerman 100 °C profile: removing it leaves the pooled LOTO RMSE unchanged (2.115 vs. 2.121) and gives the best pooled extrapolation (11.41 vs. 11.87), chiefly through the Mao batches, whose half-profiles do not identify k_d. The shrinkage prior therefore does what it was designed to do: it keeps k_d at the prior level wherever a profile shows no late slow-down (Table 7), so that PPK-FIP with and without the term predict alike, and the term is available for the profiles, such as 100 °C, that call for it. Removing both mechanisms gives 2.277, i.e. essentially the processivity effect. (3) Removing the fingerprint prior (joint estimation with the same model averaging) gives 2.177, and with BIC selection instead of averaging 2.232; the heat map confines the loss to the sparse Åkerman column (ratios 1.24 and 1.53 at 70 °C, 1.11 and 1.16 at 100 °C), where nine or seven sampling times cannot pin down selectivities and processivity together. Conversely, the tight prior $\lambda = 1$, which is closest to the two-step method, gives 2.209 ($p < 0.001$) with its loss concentrated in Bornadel (ratio 1.12–1.13), because a tight prior holds the selectivities near their apparent fingerprint values. With $\lambda = 3$ the result is insensitive to the processivity shrinkage s_p (fixed $s_p = 0.05$ or 0.2 for all cases both give the meta-selected pooled RMSE

Table 10: Ablation of PPK-FIP (LOTO RMSE per case and pooled, molar %; p: Wilcoxon signed-rank test on paired squared errors against the full method; last column: pooled forward-extrapolation RMSE). Every variant is re-estimated in every fold.

Variant	Åk70	Åk100	Born.	Tao.25	Tao.35/.45	Mao	LOTO all	p vs full	Fcst all
PPK-FIP (full)	1.886	3.505	1.054	1.358	1.495	3.098	**2.121**	–	11.87
simplified F only (no averaging)	1.871	3.199	1.054	1.795	1.535	3.429	2.271	<0.001	11.92
full F only (no averaging)	1.815	3.514	1.054	1.316	1.465	3.098	2.109	0.002	11.89
no fingerprint prior (joint, BMA)	2.342	3.882	1.052	1.310	1.459	3.104	2.177	0.866	11.58
no prior, BIC selection	2.895	4.059	1.052	1.289	1.440	3.099	2.232	0.677	11.49
two-step, simplified (ORIG)	1.828	3.742	1.825	2.101	1.643	3.607	2.533	<0.001	12.08
two-step, full (ORIG-full)	1.655	3.293	1.825	1.440	1.591	3.314	2.312	<0.001	12.06
fixed λ=1, s_p=0.05 (no meta-selection)	1.818	3.660	1.187	1.401	1.602	3.183	2.203	<0.001	11.71
fixed λ=1, s_p=0.2 (no meta-selection)	1.818	3.742	1.179	1.397	1.601	3.183	2.209	<0.001	11.70
fixed λ=3, s_p=0.05 (no meta-selection)	1.886	3.505	1.054	1.354	1.493	3.099	2.121	0.042	11.89
fixed λ=3, s_p=0.2 (no meta-selection)	1.884	3.505	1.050	1.358	1.495	3.098	2.121	0.983	11.84
no pseudo-processivity	1.867	3.906	1.718	1.512	1.619	3.108	2.289	<0.001	12.34
no deactivation	1.847	3.506	1.052	1.317	1.497	3.094	2.115	<0.001	11.41
no processivity, no deactivation	1.813	3.887	1.716	1.439	1.617	3.105	2.277	<0.001	11.99

to three decimals), and meta-validation chose $\lambda = 3$ for every case. (4) Model averaging over the simplified and full sets of lumped constants is not essential: the full set alone performs similarly (2.109), whereas the simplified set alone is worse (2.271, ratio 1.32 for Tao a_w 0.25). The two orange rows of the heat map, finally, show the original two-step forms red in almost every cell, with ratios of 1.73 for Bornadel and 1.55 for Tao a_w 0.25.

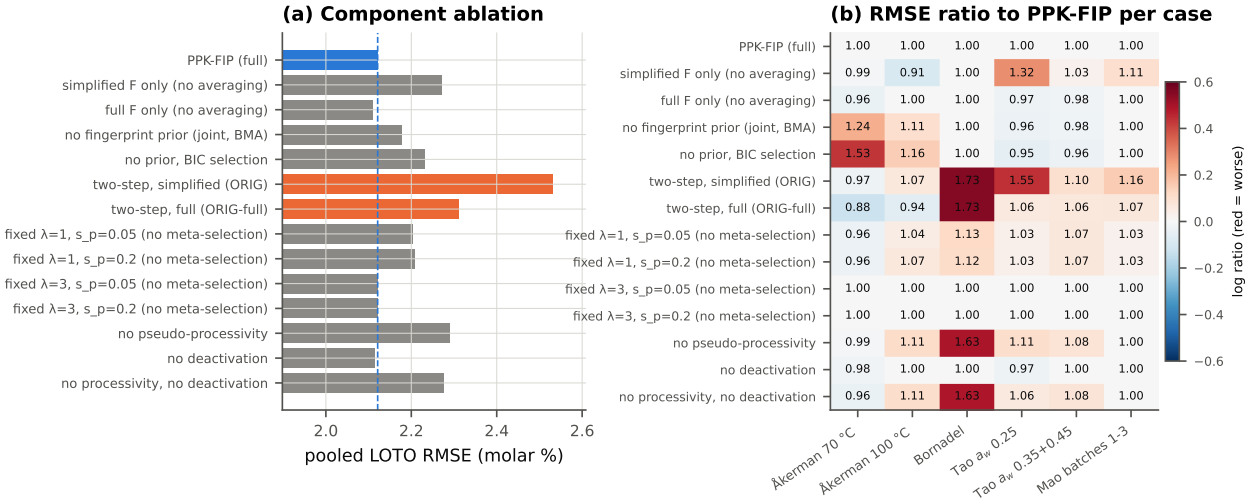


Figure 13: Ablation (LOTO). (a) Pooled RMSE of each variant; dashed line: full PPK-FIP. (b) Per-case RMSE ratio to PPK-FIP (red: worse than PPK-FIP; blue: better).

A multi-task extension tests the opposite direction, adding rather than removing structure (Table 11, Figure 14). Profiles of the same reaction system share the composition path (L1), additionally the lumped constants with hierarchical shrinkage (L1h), or everything (L2), in the spirit of the original fingerprint step, which already pooled these profiles. Full pooling strongly improves Åkerman 100 °C (3.505→2.203) and both Tao cases (1.36→1.26, 1.50→1.30) but degrades Åkerman 70 °C (1.89→2.43), because the two temperatures share a path but not the deactivation; with the pooling level chosen by leave-one-case-out meta-validation the pooled RMSE is 2.117 versus 2.121 without pooling, i.e. no net gain, and pooling was not adopted.

Role in the argument. The ablation turns a comparison into an attribution. Of the four components of PPK-FIP, pseudo-processivity and the calibrated prior are necessary; deactivation is a profile-specific option that the prior switches off where it is not needed; and model averaging can be replaced by the full lumped-constant set without loss. The dose–response along the prior width, from $\lambda = 1$ (2.21) through $\lambda = 3$ (2.12) to a flat prior (2.18), is the empirical answer to the question posed in the Introduction: the fingerprint should be used as information of calibrated width, neither as a constraint nor discarded. Pooling shows that sharing the path across conditions, the fingerprint idea in another guise, pays where the conditions differ only in time-scale (Tao), whereas per-profile estimation is preferable where they also differ in mechanism (Åkerman).

Table 11: Multi-task extension (LOTO RMSE, molar %): profiles of the same system share the composition path (L1), additionally the lumped constants with hierarchical shrinkage ($\tau = \ln 3$, L1h) or fully (L2). The level is chosen by leave-one-case-out meta-validation; the last row gives the pooled RMSE of the meta-selected levels (last column) versus L0.

Case	ORIG	ORIG-full	L0 (PPK-FIP)	L1 path	L1h hier.	L2 path+F	meta-choice
Åkerman 70 °C	1.828	1.655	1.886	2.182	2.187	2.430	L2
Åkerman 100 °C	3.742	3.293	3.505	3.325	3.002	2.203	L0
Bornadel	1.825	1.825	1.054	–	–	–	L0
Tao a_w=0.25	2.101	1.440	1.358	1.414	1.433	1.262	L2
Tao a_w=0.35+0.45	1.643	1.591	1.495	1.332	1.350	1.301	L2
Mao batches 1–3	3.607	3.314	3.098	–	–	–	L0
All cases (pooled)	2.533	2.312	2.121				2.117

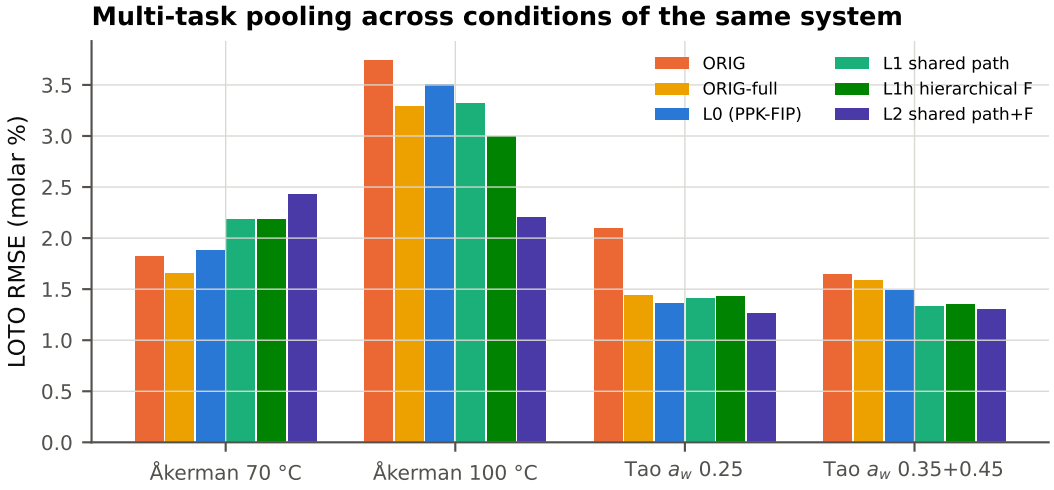


Figure 14: Multi-task pooling across conditions of the same system (LOTO RMSE).

3.5 Step 5: The Added Mechanisms Behave Like Properties of the Reaction Systems

If processivity is a property of the reaction system rather than a fitting device, a composition path learned at some conditions should transfer to others. Table 12 and Figure 15a report the RMSE of each held-out profile predicted against c from the other profiles of the same system, with the original path ($p_M = p_D = 0$) and with the processive path. The processive path was better for all three Tao water activities (by 11–31 %), equal for Åkerman 70 °C and the three Mao batches, and for Åkerman 100 °C (2.26 vs. 2.63) it shows the boundary of what a c-domain path can carry; overall the RMSE ratio was 0.975. Figure 15b shows the Tao a_w 0.45 profile predicted from a_w 0.25 and 0.35: the original path over-predicts M below $c = 100$ and under-predicts D, the same signature as in Bornadel on a smaller scale, and the processive path removes it. The Tao processivity is small ($p_D \approx 0.05$–0.06) but consistent across the three water activities (Table 7), which is why it transfers; the two Åkerman temperatures differ in deactivation, a time-domain property that the c-domain path, by construction, does not carry and that PPK-FIP estimates per profile instead.

The Monte-Carlo experiment answers the question with known truth. Table 13 and Figure 16 summarise 60 synthetic replicates per scenario; the upper row of Figure 16 shows the RMSE of the fitted curves against the noise-free truth and the lower row the log-bias of $\hat{S}_M$, for six estimators and three truths. When the truth is the original model (S1), PPK-FIP and the original method recover the true curves equally well (0.667 vs. 0.638 molar %, within one standard deviation), PPK-FIP's S_M is unbiased (+0.025 against +0.067 for the two-step method), and the unregularised joint fit over-fits (0.794, with an S_M spread four times wider): the fingerprint-informed prior is what prevents spurious complexity. When the truth is pseudo-processive (S2), the original method is biased in a way no amount of data can cure: $\hat{S}_M$ is too high by 76 % (ln-bias +0.57) with a curve error of 1.213, the tight prior $\lambda = 1$ retains half of that bias (+0.22), and PPK-FIP recovers the truth with a bias of +0.05 and a curve error of 0.260. With within-batch deactivation (S3) the curve error falls from 1.739 (original) and 0.943 (original full) to 0.718; the full F set can mimic a slow-down but only at the price of a biased S_D (−0.19), while the variants with an explicit k_d are unbiased.

Role in the argument. Step 5 closes the loop on the "apparent selectivities" of the Introduction. With

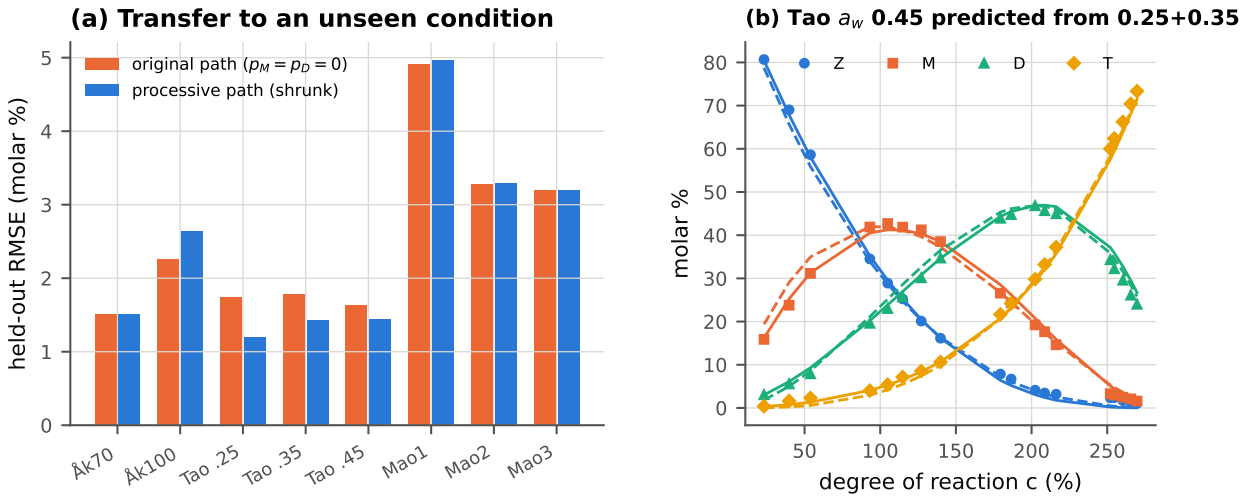


Figure 15: (a) Cross-condition transfer of the composition path. (b) Example: the profile at $a_w = 0.45$ predicted from the profiles at $a_w = 0.25$ and 0.35 (dashed: original path; solid: processive path).

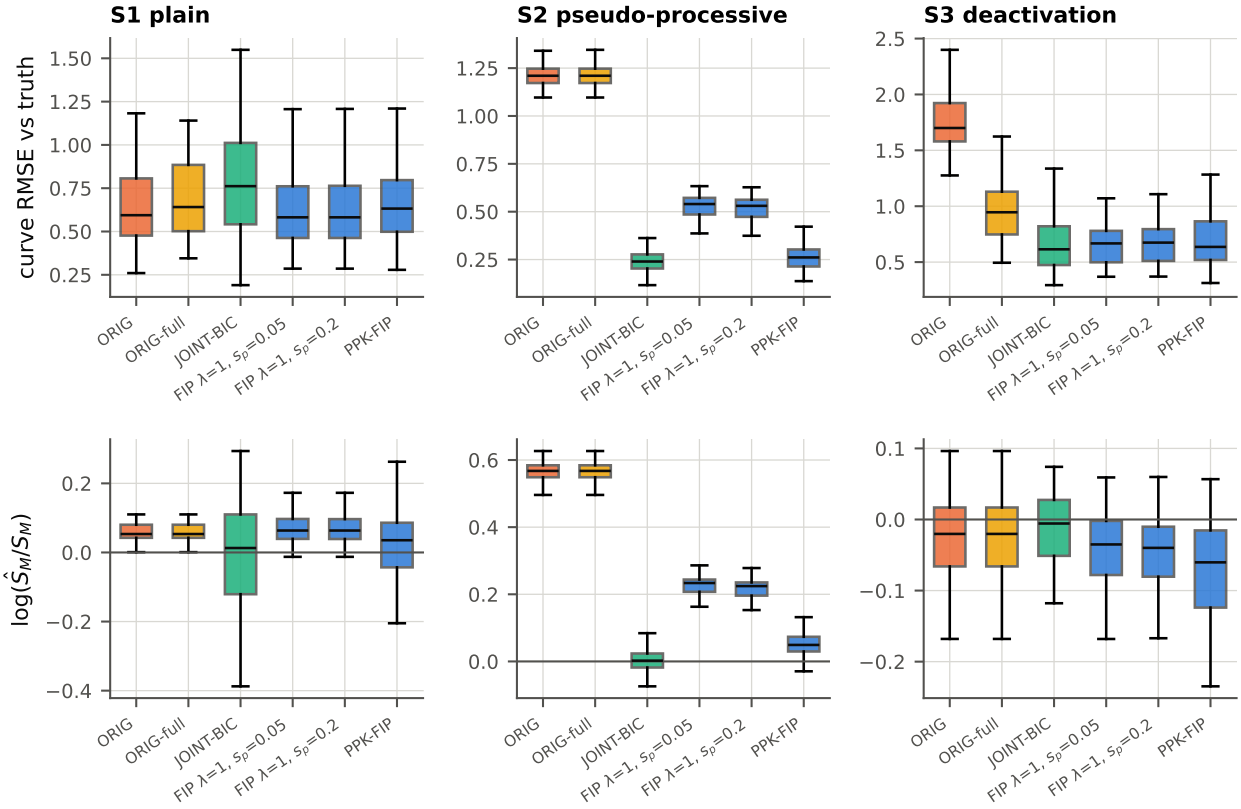


Figure 16: Monte-Carlo experiment: curve RMSE against the noise-free truth (top) and log-bias of $\hat{S}_M$ (bottom) for three truths. PPK-FIP uses the meta-selected $\lambda = 3$, $s_p = 0.2$; FIP with $\lambda = 1$ shown for comparison.

Table 12: Cross-condition transfer of the composition path: RMSE (molar %) for the held-out profile predicted against c from the other profiles of the same system.

Held-out profile	Original path	Processive path	Ratio
Akerman100	2.260	2.634	1.166
Akerman70	1.513	1.510	0.998
Mao1	4.914	4.963	1.010
Mao2	3.280	3.292	1.004
Mao3	3.196	3.196	1.000
Tao025	1.739	1.193	0.686
Tao035	1.783	1.430	0.802
Tao045	1.626	1.448	0.890
All	2.664	2.597	0.975

Table 13: Monte-Carlo truth recovery (60 replicates per scenario; mean ± SD). Curve RMSE is computed against the noise-free truth on a dense time grid.

Scenario (truth)	Estimator	curve RMSE vs truth	bias $\ln(\hat{S}_M/S_M)$	bias $\ln(\hat{S}_D/S_D)$
S1 plain	ORIG	0.638 ± 0.218	+0.067 ± 0.044	+0.147 ± 0.079
	ORIG-full	0.676 ± 0.216	+0.067 ± 0.044	+0.147 ± 0.079
	JOINT-BIC	0.794 ± 0.350	-0.013 ± 0.176	-0.110 ± 0.604
	FIP λ=1,s_p=0.05	0.637 ± 0.228	+0.071 ± 0.050	+0.076 ± 0.090
	FIP λ=1,s_p=0.2	0.639 ± 0.229	+0.071 ± 0.050	+0.076 ± 0.090
	PPK-FIP (λ=3,s_p=0.2)	0.667 ± 0.240	+0.025 ± 0.096	-0.024 ± 0.147
S2 pseudo-processive	ORIG	1.213 ± 0.060	+0.568 ± 0.028	+0.610 ± 0.046
	ORIG-full	1.213 ± 0.060	+0.568 ± 0.028	+0.610 ± 0.046
	JOINT-BIC	0.240 ± 0.054	+0.007 ± 0.041	-0.017 ± 0.097
	FIP λ=1,s_p=0.05	0.528 ± 0.069	+0.227 ± 0.031	+0.330 ± 0.051
	FIP λ=1,s_p=0.2	0.517 ± 0.071	+0.218 ± 0.031	+0.322 ± 0.053
	PPK-FIP (λ=3,s_p=0.2)	0.260 ± 0.059	+0.051 ± 0.037	+0.068 ± 0.085
S3 deactivation	ORIG	1.739 ± 0.233	-0.030 ± 0.063	-0.190 ± 0.086
	ORIG-full	0.943 ± 0.279	-0.030 ± 0.063	-0.190 ± 0.086
	JOINT-BIC	0.706 ± 0.332	-0.036 ± 0.112	-0.063 ± 0.174
	FIP λ=1,s_p=0.05	0.685 ± 0.231	-0.045 ± 0.054	-0.135 ± 0.080
	FIP λ=1,s_p=0.2	0.695 ± 0.234	-0.052 ± 0.058	-0.142 ± 0.082
	PPK-FIP (λ=3,s_p=0.2)	0.718 ± 0.267	-0.074 ± 0.079	-0.129 ± 0.109

known truth, the two-step estimator overstates S_M by a factor of 1.8 when pseudo-processivity is present, the same order as the shift from 0.909 to 0.56 found in the Bornadel data, and PPK-FIP recovers it; when the truth is the original model, PPK-FIP neither invents processivity nor loses accuracy. A generalisation of a published model must pass both tests. The transfer experiment adds real-data support that the processive path is a system property in the Tao case, and delineates the scope of the c-domain idea: it carries what is independent of the time-scale, while time-domain mechanisms such as deactivation are estimated per profile.

3.6 Step 6: Mechanism Is Still Necessary, and Machine Learning Is Most Useful on Top of It

Tables 14 and 15 and Figure 17 compare the mechanistic estimators with four data-driven models given the same information (time, condition descriptor, initial state). In leave-one-time-point-out validation (Figure 17a, log axis) the picture splits by sampling density. On the sparse Åkerman profiles all black-box models fail, with RMSE 13–21 for the Gaussian process, the MLP ensemble and TabICL and 3–4 for the neural ODE, against 1.9–3.5 for the mechanistic models. On the densely sampled profiles the Gaussian process (Bornadel 0.68, Tao a_w 0.35/0.45 0.74, Mao 1.35) and the neural ODE (1.08, 1.46, 1.53) interpolate as well as or better than any mechanistic model, and because the 141 Mao observations dominate the pooled count, the neural ODE's pooled RMSE (1.79) lies below that of PPK-FIP (2.121) even though PPK-FIP is lower on every sparse profile. The Mao result is the one anticipated in Step 2: a model that is free to leave mass conservation can follow the batch-1 plateau, whereas a mechanistic model reproduces the kinetics and leaves that feature to the measurement. In forward extrapolation (Figure 17b) every black-box model fails in every case, with per-case errors of 6–47 and pooled RMSE of 28.05 (GP), 40.79 (MLP ensemble), 24.19 (TabICL) and 20.92 (neural ODE) versus 11.874 for PPK-FIP, because nothing in them encodes the conservation of TMP or the sequential

Table 14: Leave-one-time-point-out cross-validation: mechanistic models versus black-box machine-learning baselines (RMSE, molar %).

Case	ORIG	PPK-FIP	GP	MLP ens.	TabICL	Neural ODE
Åkerman 70 °C	1.828	1.886	13.284	20.271	12.711	3.057
Åkerman 100 °C	3.742	3.505	15.073	17.701	20.607	4.064
Bornadel	1.825	1.054	0.679	12.653	1.289	1.081
Tao a_w=0.25	2.101	1.358	7.925	12.491	3.128	1.436
Tao a_w=0.35+0.45	1.643	1.495	0.741	10.694	1.567	1.464
Mao batches 1–3	3.607	3.098	1.350	12.108	2.848	1.527
All cases (pooled)	2.533	2.121	5.668	12.988	6.130	1.791

Table 15: Forward extrapolation (train $t \leq 0.5t_{\max}$): mechanistic models versus black-box machine-learning baselines (RMSE, molar %).

Case	ORIG	PPK-FIP	GP	MLP ens.	TabICL	Neural ODE
Åkerman 70 °C	1.097	1.443	37.401	39.146	37.710	6.318
Åkerman 100 °C	10.670	5.506	22.011	35.167	21.189	19.093
Bornadel	4.044	3.971	22.154	39.561	26.979	11.125
Tao a_w=0.25	19.861	17.525	46.836	43.520	29.005	33.133
Tao a_w=0.35+0.45	15.866	15.753	27.849	41.380	23.664	26.204
Mao batches 1–3	4.654	7.765	16.073	40.512	15.865	11.178
All cases (pooled)	12.080	11.874	28.047	40.791	24.188	20.919

Ping Pong structure.

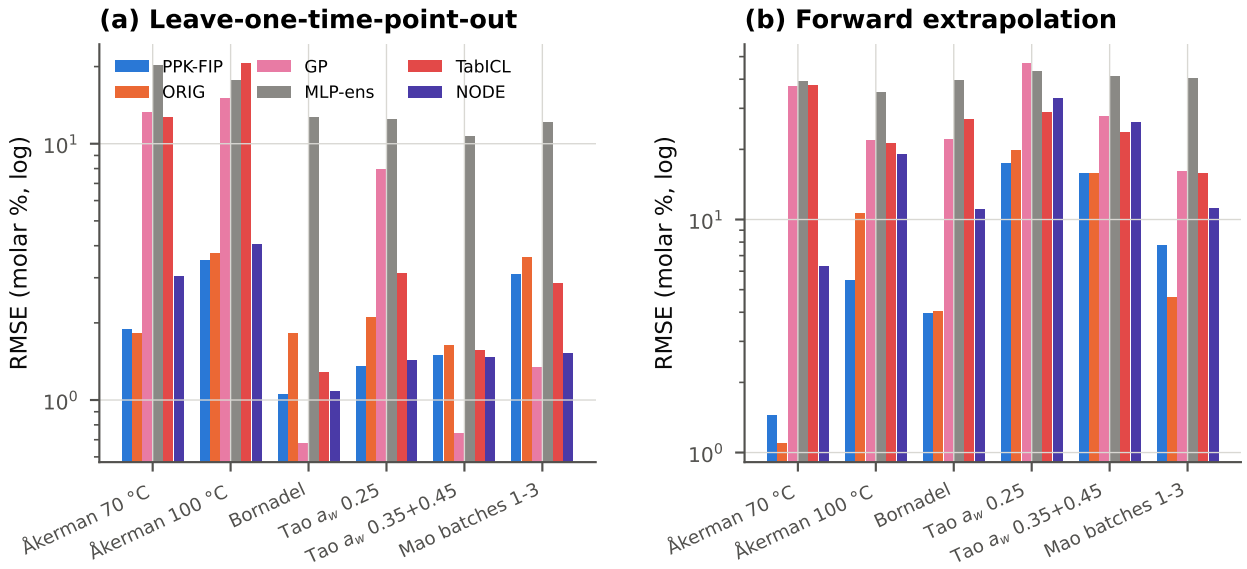


Figure 17: Mechanistic models versus black-box baselines (log scale): Gaussian process, MLP ensemble, pretrained TabICL foundation model and neural ODE.

The hybrid PPK-FIP-GP adds a Gaussian-process discrepancy to the mechanistic mean (Table 16, Figure 18). In leave-one-time-point-out validation (Figure 18a) it has the lowest error of all methods considered, 1.290 molar %, i.e. 49.1 % below the original method (2.533, $p<0.001$; lower fold error in 126 of 145 folds), below PPK-FIP itself (2.121) and below the best black-box model, the neural ODE (1.79); the gain is largest for the dense profiles (Bornadel 1.05→0.73, Mao 3.10→1.197) and negligible for the sparse ones. In extrapolation (Figure 18b) the PPK-FIP-GP bars coincide with the PPK-FIP bars (11.857 vs. 11.874), because the GP reverts to zero away from the data and the mechanistic model takes over, whereas the GP alone (pink) is off by a factor of two to thirty. Two further observations sharpen the interpretation. The same discrepancy added to the original two-step model also lowers its interpolation error (1.307, $p = 0.888$ against PPK-FIP-GP), which shows that the GP term is a general-purpose correction for digitisation and systematic error that any reasonable mechanistic mean can carry. What distinguishes the two hybrids is extrapolation: PPK-FIP-GP

Table 16: Mechanistic mean plus Gaussian-process discrepancy (RMSE, molar %). The GP is fitted to the training residuals of each species and reverts to zero away from the data.

Protocol	Case	ORIG	ORIG+GP	PPK-FIP	PPK-FIP+GP
LOTO	Åkerman 70 °C	1.828	1.778	1.886	1.839
	Åkerman 100 °C	3.742	3.546	3.505	3.486
	Bornadel	1.825	0.693	1.054	0.725
	Tao a_w=0.25	2.101	1.162	1.358	1.123
	Tao a_w=0.35+0.45	1.643	0.731	1.495	0.737
	Mao batches 1–3	3.607	1.259	3.098	1.197
	All cases (pooled)	2.533	1.307	2.121	1.290
Extrapolation	Åkerman 70 °C	1.097	1.098	1.443	1.443
	Åkerman 100 °C	10.670	10.781	5.506	5.506
	Bornadel	4.044	4.030	3.971	3.956
	Tao a_w=0.25	19.861	19.731	17.525	17.475
	Tao a_w=0.35+0.45	15.866	15.861	15.753	15.751
	Mao batches 1–3	4.654	4.308	7.765	7.733
	All cases (pooled)	12.080	12.018	11.874	11.857

inherits the mechanistic gains of PPK-FIP (Åkerman 100 °C 5.5 vs. 10.8 for ORIG+GP; pooled 11.857 vs. 12.02), so the mechanistic improvements are what count outside the data window.

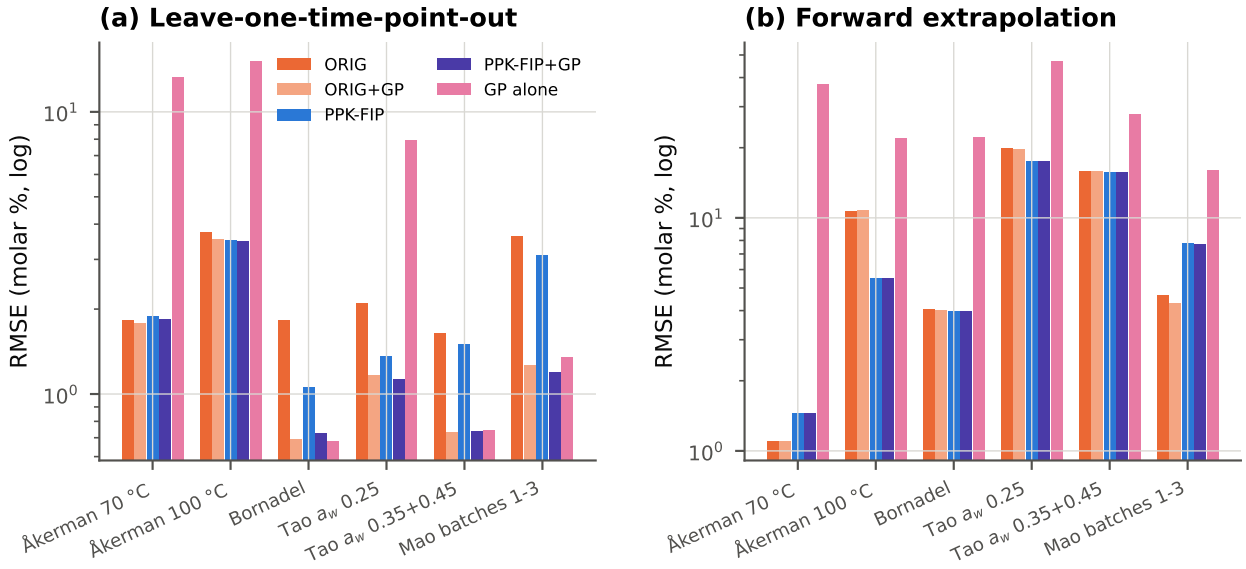


Figure 18: Mechanistic models with and without a Gaussian-process discrepancy term, and the GP alone, in (a) leave-one-time-point-out validation and (b) forward extrapolation (log scale).

Role in the argument. Step 6 answers the question left open by Step 3, whether PPK-FIP's gain is just flexibility, by separating two kinds of accuracy. Inside the experimental window a discrepancy term can patch any reasonable mechanistic mean, and interpolation accuracy says little about the mechanism; outside the window, and for the selectivities themselves, only the mechanism counts, and there black-box learners fail while PPK-FIP keeps its advantage. The practical recommendation follows: PPK-FIP for parameter estimation and extrapolation, PPK-FIP-GP when the goal is the most accurate prediction within the experimental time window.

4 Discussion

4.1 What the Chain of Evidence Establishes

The six steps of Table 1 were designed so that each result raises the question the next one answers, and read together they support one conclusion. The two-step method of Nishimura et al. is exactly reproducible and its central idea is sound, but using the fingerprint estimate as a hard constraint on an incomplete c-domain

mechanism locks a mechanistic error into the dynamic fit that in-sample statistics cannot detect (Steps 1–2). Relaxing the constraint to a calibrated prior and completing the mechanism with pseudo-processivity turns an in-sample description into an out-of-sample predictor (Step 3), the gain is attributable to exactly those two changes (Step 4), the mechanism recovers the truth where the two-step method is biased and does nothing where it is not needed (Step 5), and the mechanism, not curve-fitting flexibility, is what carries over to extrapolation and to the meaning of the parameters (Step 6). The following subsections take up the individual lessons.

4.2 What the Reproduction Audit Adds

Because Nishimura et al. published every data point, their study could be audited to a level that is rarely possible: 36 of 36 data checks and all four fingerprinting fits were reproduced exactly, and the printed parameters of 7 of the 12 dynamic fits recover the printed objective values, which also reveals the enzyme loadings hidden in the product $k_A[E]_T$ (2.0, 5.0 and 15.0 % w/w) and the actual initial acid:TMP ratio of the Mao case (10:3.3 rather than the nominal 3:1). The audit also shows two methodological problems that are independent of our new model. First, the full dynamic models are not identifiable: the profile of F_{obj} along F_A decreases monotonically towards $F_A \to \infty$ (Figure 5b), the limit in which the rate becomes independent of acid saturation and k_E and the F constants can only be estimated as ratios [19, 29]; the very large printed confidence intervals of the full models are a symptom of this ridge, and so is the erratic extrapolation of ORIG-full in Step 3. Second, several printed full-model fits are not at the minimum of their own objective (e.g. Åkerman 100 °C: printed 117.9, global minimum 48.9), and random multistarts reach the global minimum in only 46–94 % of starts (Figure 5a). Since the choice between the full and the simplified model was made by comparing BIC values of fits that were not all at their optimum, some model-selection conclusions of the original paper (for example that the simplified Mao model has a higher BIC than the full model) should be re-read with the global optima of Table 5. For the Tao case study the printed dynamic fits could not be reproduced from the published data under any initial acid ratio; we therefore use our own re-fits of the original method as the reference throughout, which is the stronger (lower-error) baseline. The general lesson is the one that shaped our evaluation design: an in-sample F_{obj} or BIC, especially of a non-identifiable model, is not evidence that a kinetic model will predict, and kinetic parameters should be accompanied by out-of-sample validation.

4.3 Why Pseudo-Processivity Matters, and Where Deactivation Applies

Nishimura et al. state that the data of Bornadel et al. are affected by mass-transfer limitation and that their selectivities are therefore only apparent, but kept the irreversible model "in the interests of simplicity". PPK-FIP makes this choice data-driven. For the Bornadel profile the data select pseudo-processivity decisively (Step 2), the probability that a freshly formed monoester is re-acylated before release ($p_M \approx 0.18$) is consistent with the processive fingerprint analysis of Mitchell et al. [13], and the residual pattern it removes is one that no lumped constant can remove, because it is fixed by the composition path. Ignoring it does not merely inflate F_{obj}: in the Monte-Carlo experiment with a pseudo-processive truth the two-step estimator of S_M is biased by +0.57 on the log scale, whereas PPK-FIP is nearly unbiased (Step 5), and in cross-validation the processivity term alone accounts for most of the out-of-sample gain (Step 4). Conversely, when the truth is the original model, the shrinkage prior and BIC weighting keep PPK-FIP from inventing processivity (scenario S1), which is the property a generalisation of a published model must have. First-order deactivation within a batch is a profile-specific mechanism. It is selected where the late part of the profile slows down more than saturation can explain, at Åkerman 100 °C ($k_d = 0.018\,\mathrm{h}^{-1}$, well above the usual operating temperature of the immobilised lipase), and stays at the prior level ($k_d \approx 10^{-3}\,\mathrm{h}^{-1}$, i.e. no deactivation within the observed time) at 70 °C and in the Tao and Bornadel profiles. Because the shrinkage prior switches it off where it is not needed, PPK-FIP with and without the term predict alike in cross-validation (Step 4), so the term can be carried at no cost and is available for the conditions, higher temperatures and longer or repeated runs, in which it is expected to matter.

4.4 The Two-Step Method as a Limiting Prior

Expressing the fingerprint step as a prior clarifies what the two-step method assumes. Fixing S_M and S_D is optimal only if the fingerprint estimate is exact; the time-domain data then cannot correct it even when they disagree, as they do in the Bornadel case, where the fingerprint fit has a sum of squares 2.5 times larger than the dynamic fit. A flat prior (the joint fit) uses no fingerprint information at all and over-fits small profiles, as the sparse Åkerman column of the ablation and scenario S1 of the Monte-Carlo experiment show. The calibrated prior of PPK-FIP lies between these limits, and the data confirm that the intermediate choice is best: leave-one-case-out meta-validation selected a prior three times wider than the fingerprint confidence interval ($\lambda = 3$) for all six cases, and in the ablation both the tighter prior closest to the two-step method

($\lambda = 1$, LOTO RMSE 2.209) and the flat prior of the joint fit (2.177) predict worse than PPK-FIP (2.121). The fingerprint information is valuable and should be kept, but as information, not as a constraint. This supports the central idea of Nishimura et al. while removing its main limitation, and it also explains why the multi-task pooling of Step 4 brought no net gain: sharing the composition path across conditions is the fingerprint idea in another guise, and it pays where the conditions change only the time-scale (Tao), whereas per-profile estimation is preferable where they also change the mechanism (Åkerman).

4.5 Mechanistic Structure versus Black-Box Learning

Modern data-driven models are not a substitute for kinetic structure in this setting. With 7–30 sampling times per profile, the Gaussian process, MLP ensemble, pretrained TabICL foundation model and neural ODE all interpolate reasonably inside the observed range of densely sampled profiles, and the neural ODE interpolates the Mao batches more closely by following the batch-1 plateau that mass conservation excludes. All of them fail in extrapolation (Step 6), because nothing in them encodes mass conservation, the Ping Pong rate law or the sequential nature of the esterifications; the neural ODE's prediction that the monoester re-appears late in the Bornadel run (Figure 12) illustrates what is lost. The improvements of PPK-FIP come instead from the mechanism the data actually require and from better use of the fingerprint information, and the exact time reparameterisation is what makes global optimisation, hundreds of cross-validation refits and Monte-Carlo studies cheap enough to do routinely. Machine learning is most useful *on top of* the mechanistic model: a Gaussian-process discrepancy absorbs the systematic and digitisation errors that no kinetic model should try to explain, giving the lowest interpolation error of all methods (PPK-FIP-GP), while the mechanistic mean guarantees sensible behaviour outside the data. That the same discrepancy term also rescues the interpolation of the original two-step model, but not its extrapolation, is the clearest demonstration in the paper that interpolation accuracy and mechanistic adequacy are different things.

4.6 Implications for Biolubricant Process Design

For the design of processes that convert biomass-derived fatty acids and TMP into biolubricants, the results have three practical consequences. Apparent selectivities obtained in a mass-transfer-limited reactor describe that reactor, not the enzyme: the Bornadel values ($S_M = 0.909$) would predict a different monoester-to-diester balance from the processivity-corrected values ($S_M = 0.56$) if transferred to a reactor with better mixing or smaller particles, so a processivity correction should accompany any such transfer. Within a system whose conditions change only the time-scale, the composition path can be transferred between conditions (Tao water activities), which reduces the experimental effort needed to characterise a new condition to the estimation of the rate constants. And kinetic parameters intended for design should be reported together with an out-of-sample check such as leave-one-time-point-out cross-validation, which here cost seconds per fold with the exact reparameterisation and which separates models that describe from models that predict.

4.7 Scope of the Results

The results should be read within the following scope. All profiles were digitised from figures of the source publications, so the residual level includes digitisation error and differs between studies, and the Mao batch-1 plateau is treated as a measurement feature rather than as kinetics; this is the kind of error that the Gaussian-process discrepancy of PPK-FIP-GP is designed to absorb. The statistical comparisons rest on leave-one-time-point-out validation with 145 folds and 533 observations; the forward-extrapolation protocol, with one split per case, characterises behaviour beyond the data window and the information content of half-profiles rather than ranking methods. The lumped constants remain empirical, as the original authors emphasised, and the full set is determined only up to the ratios of Figure 5b, so interpretation should focus on the predictions and on the identifiable parameters of the composition path (S_M, S_D, p_M, p_D) and of deactivation (k_d). Pseudo-processivity is an effective description of mass-transfer effects; its parameters belong to the reactor and agitation conditions and should be re-estimated when these change, which the fingerprint prior and the exact reparameterisation make inexpensive. Finally, the prior width was meta-validated on the six case studies of this reaction class; the same procedure transfers directly to a new class of lipase-catalysed process, where λ is selected from that class's own cases.

5 Conclusions

This paper followed one argument through six steps on the public data of Nishimura et al. (1) The two-step semi-mechanistic approach is transparent and reproducible from its published data to the last digit for the data checks, the fingerprint step and the simplified dynamic fits; but several printed full-model fits are not at

their optimum, the full model is structurally non-identifiable, and the Tao case study cannot be reproduced, so in-sample fit statistics are not a sufficient basis for model selection. (2) The misfit the method leaves has a mechanistic signature: pseudo-processivity in the mass-transfer-limited Bornadel data and, specific to a single profile, within-batch deactivation at 100 °C. (3) PPK-FIP keeps the Ping Pong core and the idea that selectivities are best learned against the degree of reaction, but uses the fingerprint estimate as prior information, adds the two mechanisms as testable options, and is fitted globally with an exact, fast time reparameterisation; it fits all six case studies better and predicts held-out observations with 16.2 % lower error than the original method and 8.2 % lower than its full variant. (4) Pseudo-processivity and the calibrated prior carry this gain; deactivation is a profile-specific option that the prior activates only where a profile slows down late, and the full lumped-constant set can replace model averaging. (5) With known truth, the two-step method overestimates the selectivity S_M by 76 % when pseudo-processivity is present, whereas PPK-FIP is nearly unbiased and reduces to the original model when processivity is absent; the processive path transfers across the Tao water activities. (6) Black-box learners interpolate dense profiles but fail in extrapolation; combined with a Gaussian-process discrepancy (PPK-FIP-GP), the mechanistic model halves the cross-validated error of the original method and outperforms every black-box model while extrapolating like the mechanism.

For the design of biolubricant processes from biomass-derived fatty acids and TMP this means that apparent selectivities obtained in mass-transfer-limited reactors should not be transferred to other reactor configurations without a processivity correction, that the fingerprint idea is most valuable as prior information rather than as a constraint, and that kinetic parameters should be reported with out-of-sample validation rather than in-sample fit statistics alone.

Data and Code Availability

All data are those published in the Supplementary Material of [10]. The code that parses them, reproduces the original analysis, implements PPK-FIP and generates every number, table and figure of this paper is provided with the manuscript (directory `code/`).

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